

Surreal Fields Across Set-Theoretic Universes

Omitted cuts, fresh-sign gaps and saturation spectra of old surreal fields; first new
birthdays; surcomplex real forms;
and a Boolean completion of reverse simplicity

22 September 2026

Abstract

Let $M \subseteq N$ be transitive universes of ZFC with the same ordinals, and view the old surreal field $\mathbf{No}^M$ as a proper class of N . All sizes, types and parameter sets are computed in N . For every uncountable cardinal κ of N , the ordered field $\mathbf{No}^M$ is κ -saturated exactly when N has no new ordinal-valued sequences of length less than κ . The same condition is equivalent to interpolation of every set-sized cut with just one side of size less than κ . The least length δ of a new sequence is a regular cardinal of both universes. A fresh sequence of length δ yields an omitted gap of exact cofinality pair (δ, δ) , by three explicit constructions: a nested interval tree, an absolute interval code with all ordinals as labels, and ordinal block codes of signs.

Every gap of $\mathbf{No}^M$ with set-sized cofinal and coinital witnesses comes from a unique fresh sign, a new sign sequence all of whose proper prefixes are old. Its fillers are exactly the sign extensions of that fresh sign, and every such gap is symmetric. The first new birthday β satisfies $\delta \leq \text{cf}^N \beta \leq \beta$. At the countable boundary, ω -saturation detects exactly the new reals. The pure field $\mathbf{No}^M[i]$ stays saturated over every N -set, while naming conjugation restores the real spectrum. The old field is closed and nowhere dense in $\mathbf{No}^N$, and each N -set of old numbers is uniformly discrete. We also treat bounded birthday fragments and the Cohen, $\text{Add}(\kappa, 1)$ and Prikry extensions. Under Global Choice, we give nonconjugate real forms of $\mathbf{No}^N[i]$ none of which has a set-sized cofinal subset. Finally we give a set-complete class Boolean algebra in which reverse simplicity is dense; class-completeness is impossible under Global Choice.

Status and provenance. An AI-assisted research report merged from four manuscripts of 22 September 2026 (sources 01, 03, 04 and 08 of the batch that placed them), each pinned to repository commit 4f26455; see section 1.6. The principal results are proposed contributions with written proofs, not certified priority claims. The report has not been refereed, and no statement in it is Lean-verified. Classical surreal facts, quantifier elimination, standard forcing facts, and the class back-and-forth and conjugacy criteria already in the collection are credited, not claimed. The finite Python checks test finite sign and interval formulas only.

Contents

1	The question, the answers, and how this report was assembled	1
1.1	The question	1
1.2	Standing hypotheses and the two levels of assumptions	1
1.3	Conventions and notation	2
1.4	The main results	2
1.5	What is offered as new, and what is not	4
1.6	Four manuscripts, one report	4
1.7	Relation to the collection	6
I	Foundations and the common spine	7
2	Foundations, notation, and imported facts	7
2.1	The two universes	7
2.2	Signs, birthdays and cones	7
2.3	Field theory and types	8
3	Absoluteness and ground-set bounds	9
4	The first new sequence length	11
5	Coding a fresh sequence by an omitted cut	12
5.1	Main route: an arbitrarily branching interval tree (source 01)	12
5.2	Second route A: an absolute all-ordinal interval code (source 04)	13
5.3	Second route B: ordinal block codes of signs (source 08)	14
6	The exact spectrum and its first gap	16
6.1	Proof of the main equivalence	16
6.2	Exact two-sided cofinalities	16
7	The countable boundary: no new reals is weaker	17
II	Fresh signs classify the set-presented gaps (source 03)	18
8	The gap–fresh-sign correspondence	18
9	The least gap	20
III	The first new birthday (source 08)	20
10	First new birthdays and their relation to saturation	21
IV	Surcomplex fields, fragments, forcing and topology	22
11	Surcomplex numbers: a language-dependent dichotomy	22
11.1	The pure field forgets the sequence obstruction	22
11.2	Naming conjugation restores the old ordered field	23
11.3	A nonisomorphism consequence	24
12	A sharp theorem for bounded birthday fields (source 01)	24

13 Forcing examples and distributivity	25
13.1 The forcing-theoretic formulation	25
13.2 A Cohen real destroys even finite-parameter saturation	26
13.3 Adding a subset of a regular cardinal	26
13.4 Prikry forcing separates the two countable notions	27
13.5 Comparison table	28
14 Why order gaps do not become Cauchy limits	28
 V Class isomorphisms and real forms (Global Choice)	 30
15 Pure-field class isomorphisms	30
16 Gap invariants of surcomplex real forms	31
16.1 Several ground real forms in one field	33
 VI A Boolean completion of reverse simplicity (source 01)	 34
17 A Boolean completion of reverse simplicity	34
17.1 A direct limit with set-coded elements	34
17.2 Why joins of all subclasses are impossible	35
 VII Scope, questions and limits	 35
18 Scope and significance	36
18.1 What the new information consists of	36
18.2 Limits of the assertions	36
18.3 A formalization route without class-sized type objects	36
19 Questions, and which of them the merge answered	37
20 What this report does not claim	38
 A Provenance and repository comparison	 42
A.1 Pinned review scope of the sources	42
A.2 Statements corrected against the current collection	43
A.3 Literature boundary	43
 B Dependency and proof-obligation ledger	 44
 C Finite checks	 45
 References	 46

1 The question, the answers, and how this report was assembled

1.1 The question

Forcing preserves ordinals, but it can add new functions on old ordinals. Consequently, it can add new surreal sign sequences without changing the ordinal labels used for birthdays. This raises a more delicate question than whether new surreal numbers exist:

How much of the old surreal field’s ability to realize cuts and first-order types survives when the surrounding universe acquires new sets?

An old field can remain an elementary subfield of a larger surreal field while losing saturation. Elementarity tests individual first-order formulas with finitely many parameters. Saturation also tests whole sets of formulas, and those sets can be new even when every formula and every parameter is old.

The familiar cut principle says that every pair of *sets* $A < B$ of surreals has a separator. The word “set” hides a parameter: which universe decides what counts as a set? When M is enlarged to N without changing the ordinals, every old surreal number remains available, but a new set of old numbers need not have belonged to M . The old cut principle promises nothing for such a collection. This report measures exactly what is lost.

1.2 Standing hypotheses and the two levels of assumptions

- (H) $N \models \text{ZFC}$, and M is a parameter-definable transitive inner class of N that contains every ordinal of N and satisfies ZFC (as a scheme). Equivalently for our purposes, M may be an available class predicate in a class theory that has the Separation and Replacement instances for that predicate, or a two-universe metatheory with these closure properties stipulated. Forcing extensions $N = M[G]$ are the intended examples, but no proof uses forcing except in section 13 and corollary 16.7.
- (GC) In addition, the ambient theory is GBC with Global Choice, and M is a class of that class universe; M need not be definable.

Everything in this report is proved under (H) except the following, which also use (GC): the class isomorphisms and real forms of part V, and the impossibility of a class-complete Boolean algebra (theorem 17.3). Sources 03 and 08 were written with (GC) as their ambient theory. Their own dependency statements confine Global Choice to the class back-and-forth and real-form arguments (source 03, its audit table; source 08, its foundations section). Every proof taken from them for a statement outside part V was re-read for this merge. It uses only set-sized choices, Choice inside M , and Replacement for the predicate M . One re-reading was needed: source 03 quantifies over gaps of No^M as class partitions. Under (H) a gap is instead given by its set presentation (definition 6.1), and the statements of part II become schemes over such presentations.

External means computed in N , never in a larger metatheory. If M and N are represented by countable transitive *set* models inside a larger universe, all claims about sets, cardinalities and saturation must still be read inside N . They are not claims of saturation with respect to every subset in that larger universe. We use no unrestricted truth predicate for class structures. In the languages used here, quantifier elimination reduces formulas to finite algebraic conditions and so supplies the satisfaction relation directly. Types are ordinary N -sets of finite syntactic expressions over set-sized parameter sets. This avoids the general class-satisfaction obstacle discussed by van den Dries–Ehrlich and by Ehrlich [3, 6].

The Prikry examples assume a measurable cardinal in M . The $\text{Add}(\kappa, 1)$ examples need no cardinal arithmetic; source 04’s extra hypothesis $\kappa^{<\kappa} = \kappa$ is used only for its additional preservation clause (example 13.3).

1.3 Conventions and notation

The four sources used different letters for the same objects, and several letters for two objects. This report fixes one symbol per notion.

Notion	Symbol here	In the sources
Inner and outer universe	$M \subseteq N$	$M \subseteq N$ (01, 03); $M \subseteq W$ (04); $M \subseteq V$ (08)
Old and new surreal fields	$\mathbf{No}^M, \mathbf{No}^N$	K, K^+ (01); F, E (03); S, T (04); F_M, F_V (08)
Old and new surcomplex fields	$\mathbf{SC}^M = \mathbf{No}^M[i],$ $\mathbf{SC}^N = \mathbf{No}^N[i]$	C, C^+ (01); K_M, K_N (03); K_M, K_W (04); K_M, K_V (08)
Canonical conjugations	c_M, c_N	c (01); c_M, c_N (03); c_M, c_W (04); c_M, c_V (08)
Transported conjugation	$j = \Phi c_M \Phi^{-1}$	j_M (03); d (04); c' (08)
First new length of an ordinal-valued sequence	$\delta = \delta(M, N)$	δ (01, 08); ν (03); $d(M, W)$ (04)
First new birthday	$\beta = \beta(M, N)$	β (08)
No new short sequences	$\mathbf{NS}_{<\kappa}(M, N)$	same (01, 08); $\mathbf{D}_{<\kappa}(M, W)$ (04)
Birthday (length) of a sign	$\mathbf{b}(s)$	$\mathbf{b}$ (01, 04); $\ell(s)$ (03); $\mathbf{lh}(s)$ (08)
Sign cone of s	$\mathbf{Cone}(s)$	(L_s, R_s) separators (01); $\mathbf{Cone}(s)$ (03, 08); C_s (04)
Ordinal block codes	$\text{code}(f), \text{code}^+(f)$	$C(f)$ (08); s_f (03)
Measurable cardinal (Prikry)	ρ	ρ (01); λ (03); ν (04); θ (08)
Regular cardinal of $\text{Add}(\kappa, 1)$	κ	ρ in $\text{Fn}_{<\rho}(\rho, 2)$ (01); κ (03, 04, 08)
Gap with set-sized witnesses	set-presented gap	set-character gap (01); set-presented gap (03); Dedekind gap with character (08)
Cut property	η_κ	η_κ (01, 03, 04); $\text{Cut}_{<\kappa}$ (08)

The letter K is not used for the proper-class fields $\mathbf{No}^M$ and $\mathbf{SC}^M$: most reports of the collection reserve K for a *set-sized* field, and the collection’s notation guide recommends $\mathbf{SC} = \mathbf{No}[i]$ for the surcomplex class (docs/NOTATION.md). The letter δ always denotes the first new length of an ordinal-valued sequence, never a birthday; the first new birthday is β , as in source 08. The sibling report docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets uses the same letter β for the first new birthday. Several of that report’s source manuscripts wrote δ for the first new birthday; that δ is the β of this report, not its δ . The letter θ denotes an ordinal bound, on the range of a sequence or on a set of birthdays, and ρ is reserved for the measurable cardinal of the Prikry examples. The letter ν is not used.

Tempting false readings. “Gap” here always means a gap of the class order $\mathbf{No}^M$ seen from N , with set-sized witnesses; it is not the proper-class “finite–infinite gap” of $\mathbf{No}$ itself studied in docs/surreal/genetic-gaps-and-primitives. “External” means computed in the outer universe N ; in the foundations report the same word refers to a stronger metatheory (remark 14.4). η_κ is indexed by a size bound, not by an aleph subscript. $\beta(M, N)$ is an ordinal attached to a pair of universes, not the birthday of a particular number.

1.4 The main results

For an infinite cardinal κ of N , write

$$\mathbf{NS}_{<\kappa}(M, N) \iff (\lambda\theta)^N \subseteq M \quad \text{for all ordinals } \lambda < \kappa \text{ and } \theta. \quad (1.1)$$

Every ordinal-valued sequence has bounded range, so this is exactly the absence of new ordinal-valued sequences of length less than κ . The range bound θ is *not* required to be below κ .

Theorem 1.1 (Main spectrum theorem; sources 01, 03, 04, 08). *For every infinite cardinal κ of N , the following are equivalent.*

- (i) $\text{NS}_{<\kappa}(M, N)$ holds.
- (ii) Every pair of N -sets $L, R \subseteq \mathbf{No}^M$ with $L < R$ and $\min(|L|, |R|) < \kappa$ has a separator in $\mathbf{No}^M$.
- (iii) Every such pair with $|L \cup R| < \kappa$ has a separator in $\mathbf{No}^M$.

For uncountable κ , these are also equivalent to κ -saturation of $\mathbf{No}^M$ in the ordered-field language, and to κ -saturation of the order $(\mathbf{No}^M, <)$ as a dense linear order, both evaluated in N . When they fail, an omitted pair as in (iii) can be chosen inside $(0, 1)$.

Condition (ii), from source 01, is stronger in appearance than the usual small-cut condition: the other side can be an arbitrarily large set. Neither side may be a proper class. In particular a cut is always filled if one side is an actual ground-model set (theorem 3.5). The two-sided form (iii) and the saturation clause were proved independently by all four sources; the dense-linear-order clause is source 08's and the $(0, 1)$ clause source 04's.

Theorem 1.2 (First loss and language sensitivity; sources 01, 03, 04, 08). *Suppose $M \neq N$, and let δ be the least ordinal length of a new ordinal-valued sequence. Then:*

- (i) δ is an infinite regular cardinal in both M and N .
- (ii) $\mathbf{No}^M$ has a set-presented gap of exact cofinality pair (δ, δ) , and no set-presented gap has either cofinality smaller than δ .
- (iii) For uncountable N -cardinals κ , $\mathbf{No}^M$ is κ -saturated exactly when $\kappa \leq \delta$.
- (iv) $\mathbf{No}^M$ is ω -saturated exactly when $\mathbb{R}^M = \mathbb{R}^N$.
- (v) SC^M in the pure field language is saturated over every N -set of parameters. The expansion (SC^M, c_M) by $c_M(a + bi) = a - bi$ has exactly the saturation spectrum of $\mathbf{No}^M$, including at ω .

Beyond this common spine, each source contributed results that the others do not have.

- **Fresh signs classify the gaps** (source 03, part II). The set-presented gaps of $\mathbf{No}^M$ correspond bijectively to the *fresh signs*, new sign sequences all of whose proper prefixes are old. The gap of a fresh sign s has both cofinalities equal to $\text{cf}^N b(s)$, and its fillers in $\mathbf{No}^N$ are exactly the extensions of s . Hence every set-presented gap is symmetric, and $\mathbf{No}^N \setminus \mathbf{No}^M$ is the disjoint union of the fresh cones.
- **The first new birthday** (source 08, part III). The least birthday β of a new surreal is a cardinal of both universes, and $\delta \leq \text{cf}^N \beta \leq \beta$.
- **Bounded birthday fragments** (source 01, section 12). For an uncountable regular cardinal κ of M that remains a cardinal in N ,

$$(\mathbf{No}_{<\kappa})^M \text{ is } \kappa\text{-saturated in } N \iff \begin{aligned} &\text{cf}^N(\kappa) = \kappa \text{ and} \\ &({}^\alpha 2)^N = ({}^\alpha 2)^M \quad (\alpha < \kappa). \end{aligned} \quad (1.2)$$

Thus the full-class theorem and the bounded-fragment theorem test different preservation properties.

- **Topology** (sources 01, 03, 04, section 14). Every N -set of old numbers is uniformly discrete and closed, and set-indexed Cauchy nets of old numbers are eventually constant. If $M \neq N$, the old field is closed and nowhere dense in $\mathbf{No}^N$, and its subspace topology is its own order topology.
- **Real forms under Global Choice** (sources 03, 04, 08, part V). $\mathbf{SC}^M \cong \mathbf{SC}^N$ as pure fields, over any N -set of common parameters. Transporting c_M gives involutions of $\mathbf{SC}^N$ that are not conjugate to c_N , although neither fixed field has a set-sized cofinal subset. Several such involutions can be separated by the gap invariant, and they can agree with c_N on every ordinary complex number.
- **A Boolean completion** (source 01, section 17). The reverse-simplicity order on signs embeds densely in an explicit set-complete class Boolean algebra. Under Global Choice no class-complete such algebra exists.

1.5 What is offered as new, and what is not

The proposed contributions are the exact external spectrum of theorem 1.1 with its one-sided form; the three explicit codings of a fresh sequence by an omitted cut (the arbitrary-alphabet interval tree of source 01, the absolute all-ordinal interval code of source 04, and the ordinal block codes of sources 03 and 08); the exact gap threshold; the fresh-sign classification of all set-presented gaps; the separation of the first new birthday from the first saturation obstruction; the distinction between ω - and ω_1 -saturation under no-real extensions; the resulting surcomplex language comparison; the bounded-fragment criterion; the external topological statements; the real-form invariants; and the Boolean completion. Each source stated its own contributions as proposed, with priority not certified.

Their ingredients are not claimed as new: Conway’s cut theorem and sign expansions, real closedness of $\mathbf{No}$ and algebraic closedness of $\mathbf{No}[i]$ [1, 2]; quantifier elimination for real closed and algebraically closed fields [9]; saturation and homogeneous universality of surreal class models [6, 7]; standard forcing facts [10, 11, 12, 13]; the forcing notion of a fresh function [14]; and, from the collection, class back-and-forth for proper-class algebraically closed fields, the conjugacy criteria for real forms, and a real form with a countable cofinal subset [19]. Several statements are deliberately labelled consequences rather than independent discoveries. Regularity of the first new length is an elementary set-theoretic fact, not a new invariant.

The Boolean section answers a concrete question in Berenbeim’s 2020 *notes*, not a conjecture of a peer-reviewed paper. Its construction is a natural direct limit and may be folklore in class forcing; its value here is an explicit, size-correct answer, not a claim that Boolean completions were previously unavailable. General class-forcing completion theory predates those notes [16]. Berenbeim also proposes studying surreals under forcing in the final section of the same notes [15, pp. 32, 34]; this report develops one concrete direction of that program and does not settle the notes’ broader suggestions about embeddings of universes.

1.6 Four manuscripts, one report

This report is merged from four manuscripts, all dated 22 September 2026 and all written against commit 4f2645599121fa872c7104995e47f86f0382351f of VladimirReshetnikov/Surreal. The source numbers are those of the batch in which they arrived. Delivered code and data files keep those numbers as prefixes. No source manuscript is shipped.

Source	Manuscript	Contribution here
01 (base)	<i>Forcing, Omitted Cuts, and Old Surreal Fields: ordinal-sequence spectra, surcomplex conjugation, and a Boolean completion of reverse simplicity</i> (25 pages)	The foundations and spine (sections 2 to 7), the one-old-side theorem, the interval tree, the surcomplex dichotomy (section 11), bounded fragments (section 12), the forcing framework (section 13), external Cauchy rigidity, the Boolean completion (section 17), and most of section 18. LaTeX Makefile, as <code>code/01-forcing-omitted-cuts-Makefile</code> .
03	<i>Surreal Fields Across Set-Theoretic Universes: fresh gaps, exact saturation thresholds, and surcomplex real forms</i> (21 pages)	All of part II (fresh-sign gap classification, filling cones, the terminal-plus block code, the least-gap theorem); old ordinal bounds (lemma 3.3); the ordinal-bound form of the cut criterion; the $\text{Add}(\kappa, 1)$ first-gap calculation without cardinal arithmetic; the Prikry fresh sign; the proper-class ACF saturation theorem; the real-form invariant, agreement with c_N on $\mathbb{C}^N$, several ground real forms and finite forcing towers; the closed, nowhere-dense and subspace-topology statements. Files prefixed <code>03-fresh-sign-gaps-</code> .
04	<i>Surreal Fields Across Set-Theoretic Universes: exact saturation, forcing, and surcomplex real forms</i> (25 pages)	The absolute all-ordinal interval code with closed finite-branch formulas (second route A, section 5.2); the $(0, 1)$ clause; the forcing characterization for preserved uncountable cardinals; $\text{Add}(\kappa, 1)$ under $\kappa^{<\kappa} = \kappa$; the elementarily equivalent nonconjugate involutions with different spectra and the several-spectra corollary; external discreteness in the product topology, open prefix cones, and nowhere-denseness of SC^M in SC^N . Files prefixed <code>04-universe-saturation-</code> .
08	<i>Forcing and Saturation of Surreal and Surcomplex Fields: exact preservation criteria and a real-form dichotomy</i> (24 pages)	The ordinal block code with its two-bound cone identity and exact separator class (second route B, section 5.3); the dense-linear-order clause; the first new birthday and $\delta \leq \text{cf}^N \beta \leq \beta$ (part III); the Cohen, $\text{Add}(\kappa, 1)$ and Prikry values of β ; the lottery-sum caution; the non-definability of conjugation in the pure field; the saturated back-and-forth route and the isomorphism over an arbitrary old set. Files prefixed <code>08-forcing-saturation-</code> .

Why one report. All four manuscripts prove the same spine in the same setting: the cut criterion, the regular first new length, an omitted (δ, δ) gap, the ω boundary, the pure/conjugation contrast, the distributivity corollary and the Cohen and Prikry examples. Their extras are complementary. **Source 01 is the base:** its hypotheses are the weakest (plain ZFC with a definable inner model, Global Choice only for one negative Boolean result), and its main criterion is the strongest in form, since one small side suffices.

Printed once. The following results were proved by two or more sources and appear once, with all proving sources named at the statement: the absoluteness of old arithmetic and elementarity (proposition 3.1); ground-set bounds and envelopes (lemma 3.2); coding of short families (lemma 3.4); regularity of δ (proposition 4.2); cuts versus real-closed saturation (lemma 2.3); the main spectrum (theorem 1.1); the first gap (theorem 6.2); the ω criterion (theorem 7.1); pure surcomplex saturation (theorem 11.2); the conjugation spectrum (theorem 11.4); nonisomorphism (corollary 11.6); the distributivity corollary (corollary 13.1); the Cohen, $\text{Add}(\kappa, 1)$ and Prikry examples; external discreteness (proposition 14.1); closedness and nowhere-denseness (theorem 14.3); the class isomorphism (theorem 15.2); the transported real form (theorem 16.2); and the several-grounds statements (theorem 16.6).

Marked second routes. The converse direction of the spine has three genuinely different constructions. The interval tree of source 01 is printed as the main route (section 5). The absolute interval code of source 04 (section 5.2) and the ordinal block codes of sources 08 and 03 (section 5.3 and proposition 9.1) are kept as marked second routes. Two existence proofs for δ (Mostowski collapse, sources 01 and 03; least new rank, sources 04 and 08) and two class back-and-forth routes (algebraic/transcendental extension, sources 03 and 04 as in the collection; saturated elementary maps, source 08) are kept as well.

Where the merge had to choose. (a) Symbols: every renaming is listed in section 1.3, and none changes a normalization. (b) Hypotheses: statements are printed at the weakest hypotheses that any source proved them under. Where sources differ, the difference is recorded at the statement: regularity of δ in M (sources 01, 03, 08; source 04 states N only); cardinal arithmetic for $\text{Add}(\kappa, 1)$ (only source 04 assumes it); the range of κ in the cut criterion (every infinite cardinal in source 01, every infinite ordinal in source 03, uncountable in sources 04 and 08). (c) Open questions: those answered inside the cluster are recorded as answered, with the answering source (section 19). (d) Stale repository statements in the sources are corrected in section A. Text written for the merge is confined to this section, the framing remarks that relate sources and reports, remark 10.4, the question statuses, the non-claims ledger and the appendices. It adds no mathematical claim beyond the two-line observation of remark 10.4.

1.7 Relation to the collection

Surcomplex field automorphisms (docs/surcomplex/surcomplex-field-automorphisms, prefix `saut:`). Its Theorem 9.1 (`saut:thm:acfhomogeneity`) is the class back-and-forth used in part V; its Section 10 (`saut:sec:realforms`) proves that involutions are conjugate exactly when their fixed real fields are isomorphic, and constructs a real form with a countable cofinal subset (Theorem 10.2, `saut:thm:inequivalentrealform`). Its Proposition B.1 (`saut:prop:conjugacycriterion`) says that an involution is conjugate to c if and only if its fixed real field has the set-cut interpolation property. None of the four sources cites that proposition. Read in N , its easy direction already gives the non-conjugacy statements of theorems 16.2 and 16.3, because the transported fixed field is isomorphic to $\mathbf{No}^M$, which omits an N -set cut. What is new is recorded in remark 16.4. These results *continue*, but do not settle, that report’s open “real-form and continuity classifications” (its Section 14).

Foundations (docs/foundations-and-computation/foundations, prefix `found:`). Proposition 14.1 is an external version of its Theorem 12.1 (`found:thm:discrete`); see remark 14.4 for the exact relation, including its hypothesis on bounds and the passage of its Section 7.7 on external countability.

Computable surreals (docs/foundations-and-computation/computable-surreals). Its `prop:topology` is the universe-relative version of set-indexed convergence, with an index type small in the lower universe. In the present same-ordinal setting no such smallness is needed, because $\mathbf{No}^M$ is still a proper class of N ; the two statements are consistent.

Birthday cutoffs and hereditary sets (docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets), the sibling report placed in the same batch. It studies bounded birthday fields *with a birthday symbol*, their interpretation of hereditary sets, and their behaviour under outer models. It also proves, for every same-ordinal outer model, that the first new birthday is the least ordinal acquiring a new subset. This report keeps source 08’s statement and proof of that fact, because source 08 adds cardinality in N and the inequality $\delta \leq \text{cf}^N \beta$ (remark 10.5). The two reports study different structures: here the pure ordered field and its external saturation, there the birthday-expanded fields and their interpretations.

Genetic gaps and primitives (docs/surreal/genetic-gaps-and-primitives) uses “gap” for a proper-class gap of $\mathbf{No}$; the gaps here are set-presented gaps of $\mathbf{No}^M$ seen from N . The same report cites Berenbeim’s 2022 dissertation; the 2020 notes answered in section 17 are not cited elsewhere in the collection.

Part I

Foundations and the common spine

2 Foundations, notation, and imported facts

2.1 The two universes

In N , $\mathbf{No}^M$ is a proper class: it contains an old all-plus sign of every ordinal length. The superscript in $\mathbf{No}^M$ means that the sign functions, their construction and the operations are computed in M . It is not an additional set-sized model assumed to have a cardinality. We never form a set of all surreals, a proper-class support, or a class-object whose members are all class maps.

2.2 Signs, birthdays and cones

A surreal number is a sign function $s : \alpha \rightarrow \{-, +\}$ for an ordinal α ; its birthday is $b(s) = \alpha$. The empty function represents zero. At the first place where two signs differ, numerical comparison uses

$$- < \text{termination} < +.$$

Write $s \preceq t$ when s is an initial segment (prefix) of t , allowing equality. Simplicity is the proper-prefix relation, not numerical order and not merely a smaller birthday. The ordinal α is represented by the all-plus function of length α , written $\hat{\alpha}$ when confusion with field arithmetic is possible. Its surreal successor agrees with $\alpha + 1$ as an ordinal; arbitrary ordinal addition and surreal field addition must not be conflated. Concatenation $\frown$ is ordinal concatenation.

For sets L, R of surreals, $L < R$ means $\ell < r$ for all $\ell \in L$ and $r \in R$. Empty sides are allowed. A separator means a *strict* separator: $L < x < R$. The classical cut theorem supplies the unique simplest separator $\{L \mid R\}$, a prefix of every separator [1, 2, 3]. We use the birthday bound

$$b(\{L \mid R\}) \leq \sup\{b(y) + 1 : y \in L \cup R\}, \quad (2.1)$$

with the supremum of the empty set equal to zero. Put

$$\text{Cone}(s) = \{t : s \preceq t\}.$$

Lemma 2.1 (Cones are convex; the prefix characterization; sources 03, 04). (i) *The cone $\text{Cone}(s)$ is numerically convex. That is, if $s \preceq t$ and u lies numerically between s and t , then $s \preceq u$.*

(ii) *Let I be a convex class of surreals, let $s \in I$, and suppose no proper prefix of s belongs to I . Then every $x \in I$ extends s , and s is the unique simplest element of I .*

Proof. (i) Two extensions of s have the same signs before $b(s)$; anything between them cannot first disagree with s at an earlier position.

(ii) If x is a proper prefix of s , the hypothesis excludes it. If neither sequence extends the other, their maximal common prefix u is a proper prefix of s . At the first disagreement one sequence has $-$ and the other $+$, so u lies strictly between them, and convexity gives $u \in I$, a contradiction. The remaining possibility is that x extends s ; such an extension has birthday at least $b(s)$, with equality only for $x = s$. $\square$

Lemma 2.2 (The cone of a sign; source 01). *For $s : \alpha \rightarrow \{-, +\}$ put*

$$L_s = \{s \upharpoonright \gamma : \gamma < \alpha, s(\gamma) = +\}, \quad R_s = \{s \upharpoonright \gamma : \gamma < \alpha, s(\gamma) = -\}.$$

A surreal x satisfies $L_s < x < R_s$ if and only if $s \preceq x$.

Proof. If x extends s , its comparison with each prefix in the displayed sets is decided by the next sign of s , so the strict inequalities hold. Conversely, if x terminates at $\gamma < \alpha$ while agreeing up to that point, then $x = s \upharpoonright \gamma$ violates the corresponding strict inequality. If it first disagrees at $\gamma < \alpha$, compare x with the prefix $s \upharpoonright \gamma$: the opposite signs place x on the wrong side of it. These exhaust the ways in which x could fail to extend s . $\square$

The numerical bounds

$$-\widehat{\mathbf{b}(x)} \leq x \leq \widehat{\mathbf{b}(x)} \tag{2.2}$$

follow by comparing a sign with the all-minus and all-plus signs of the same length.

2.3 Field theory and types

The full surreal field is real closed, and its extension by a square root of -1 is algebraically closed. Also $\mathbf{No}_{<\lambda}$ is a real closed subfield whenever λ is an epsilon number; in particular for every uncountable cardinal λ , since such cardinals satisfy $\omega^\lambda = \lambda$ in ordinal arithmetic. We use the corrected birthday-fragment results of van den Dries–Ehrlich and their restatement by Bournez–Guilmant [3, 4, 5]. Nothing about exponential functions is needed.

We use quantifier elimination for real closed fields in $\mathcal{L}_{\text{or}} = \{0, 1, +, -, \cdot, <\}$, for algebraically closed fields of characteristic zero in $\mathcal{L}_{\text{fld}} = \{0, 1, +, -, \cdot\}$, and for dense linear orders without endpoints [9]. Embeddings between real closed ordered fields, and between algebraically closed fields of the same characteristic, are elementary. Over a real closed parameter subfield a nonalgebraic one-variable type is determined by an order cut; over an algebraically closed field one-variable types are algebraic or transcendental.

A class structure is κ -saturated (in N) if every complete one-variable type over an N -set of parameters of cardinality less than κ that is finitely satisfiable in the structure is realized there. As usual this gives the finite-variable version: realize the projection of a type in $n + 1$ variables to the first n , then realize the specialized one-type over the enlarged parameter set. All syntactic data and parameter sets at every step are sets, and satisfaction is supplied by quantifier elimination. ω -saturation concerns *finite* parameter sets; countable parameter sets are ω_1 -saturation. Even a type over the empty set can contain countably many formulas.

An ordered class H has the η_κ property if every pair of N -sets $A, B \subseteq H$ with $A < B$ and $|A \cup B|^N < \kappa$ has a strict separator in H . Empty sides are allowed. Following source 03 we allow an infinite ordinal bound κ when needed.

Lemma 2.3 (Cuts versus saturation; sources 01, 03, 04, 08). (i) *For an uncountable cardinal κ , a real closed ordered field — including the definable class fields used here — is κ -saturated if and only if its order is η_κ .*

(ii) (source 08) *For a dense linear order without endpoints, the same equivalence holds for every infinite κ .*

(iii) (source 04) *In each universe of set theory, its own full surreal field is set-saturated as an ordered field.*

Proof. (i) Saturation realizes the finitely satisfiable partial type of the inequalities of a small cut; a complete extension exists by compactness applied to the set of formulas with those parameters, inside a set-sized real closed subfield containing them.

Conversely, let A have size less than κ . The real closure F of $\mathbb{Q}(A)$ inside the field is a set of cardinality at most $\max(|A|, \aleph_0) < \kappa$; there are set many polynomials over $\mathbb{Q}(A)$ and finitely many roots of each. A complete type over A extends to one over F that is finitely satisfiable in the ambient field. If the extended type is algebraic, it is realized in F . Otherwise it declares, for every $a \in F$, exactly one of $a < X$ and $X < a$; these inequalities form a separated cut of size less than κ . A separator y realizes the entire type: every nonzero polynomial over the real closed F factors into linear factors and irreducible quadratics; the quadratics have constant sign, and the signs of the linear factors are fixed by the cut. Quantifier elimination completes the argument. Restrict back to A .

(ii) A complete one-type of a dense linear order over A either says $X = a$ for some $a \in A$ or specifies a cut of A . There is no countable field closure to add, so the argument works even at $\kappa = \omega$.

(iii) The cut theorem realizes every set-sized separated cut; apply the proof of (i) with no cardinal bound. $\square$

The uncountability hypothesis in (i) cannot be dropped: the real closure of even the empty parameter set is countable, not finite. Every dense order without endpoints has η_ω , whereas a real closed field need not be ω -saturated (section 7).

3 Absoluteness and ground-set bounds

Proposition 3.1 (Old arithmetic and elementary inclusions; sources 01, 03, 04, 08). (i) $\mathbf{No}^M = \mathbf{No}^N \cap M$ is an initial subclass of the sign tree: every prefix of an old sign is old. Birthdays and numerical comparison of old signs are absolute.

(ii) If $L, R \in M$ are sets of old surreals with $L < R$, then $\{L \mid R\}^M = \{L \mid R\}^N$.

(iii) The inclusion $\mathbf{No}^M \subseteq \mathbf{No}^N$ preserves zero, one, negation, addition, multiplication and inversion. Every ordinal is old, so $\mathbf{No}^M$ is cofinal and coinital in $\mathbf{No}^N$.

(iv) In N , $\mathbf{No}^M$ is real closed and $\mathbf{SC}^M$ is algebraically closed, and

$$\mathbf{No}^M \preccurlyeq \mathbf{No}^N \quad \text{in } \mathcal{L}_{\text{or}}, \quad \mathbf{SC}^M \preccurlyeq \mathbf{SC}^N \quad \text{in } \mathcal{L}_{\text{fld}}.$$

Proof. (i) Being an ordinal-length function into $\{-, +\}$, its domain, and first disagreement comparison are absolute for transitive models. If $s \in M$ and $\alpha \leq b(s)$, then $\alpha \in M$ and $s \upharpoonright \alpha \in M$.

(ii) Let $u = \{L \mid R\}^M$. It still separates L, R in N . The simplest separator v computed in N is a prefix of u by the simplicity theorem in N , hence old by (i), hence an available separator in M , so $u \preceq v$ by the simplicity theorem in M . Thus $u = v$ (source 08). Equivalently, every proper prefix of u fails in M to separate, witnessed by an old element of L or R , and that witness still works in N ; lemma 2.1(ii) applied to the convex class of N -separators gives the same conclusion (source 04). The argument does not minimize birthdays over old candidates, which would not be absolute since N can contain new short signs; the prefix witnesses exclude new candidates.

(iii) Conway's addition and multiplication are defined from canonical options by well-founded, set-like recursion. For old inputs the options are old, the earlier recursive values agree by induction, the option families belong to M , and (ii) identifies the resulting cuts. Every recursion needed for particular inputs has a set-sized dependency closure; no class-recursion principle is assumed. The inverse of an old nonzero number remains its inverse by uniqueness. A large all-plus or all-minus ordinal lies above or below any given new surreal.

(iv) A polynomial over $\mathbf{No}^M$ has a finite old coefficient tuple, which belongs to M . In M positive elements have square roots and odd-degree polynomials have roots, and those witnesses work in N . Thus $\mathbf{No}^M$ is

externally real closed, and the same finite-polynomial argument gives external algebraic closedness of SC^M . Quantifier elimination gives both elementary inclusions. $\square$

This proposition asserts elementarity of the *fields*. It does not assert $M \preceq N$ as models of set theory, which is generally false for a forcing extension. “Initial” is a simplicity-tree assertion: $\mathbf{No}^M$ is not a numerical interval of $\mathbf{No}^N$, and it need not contain a new surreal whose birthday is below some old birthday.

Lemma 3.2 (Every external set of old numbers is ground-bounded; sources 01, 04, 08). (i) *If A is an N -set and $A \subseteq \mathbf{No}^M$, there are an ordinal α and a ground set $B \in M$ with $A \subseteq B = (\mathbf{No}_{<\alpha})^M$. Every N -set of elements of SC^M is likewise contained in a ground set of old surcomplex numbers, and these ground sets have well-orderings in M .*

(ii) *Such an A is contained in a set-sized real closed subfield $E \in M$ of $\mathbf{No}^M$, and an N -set of elements of SC^M in a set-sized algebraically closed subfield $E[i] \in M$; these are elementary substructures. There is an old element transcendental over any such $E[i]$.*

Proof. (i) In N form $\alpha = \sup\{b(a) + 1 : a \in A\}$. It belongs to M because the universes have the same ordinals. The set of old signs of length below α exists in M by Power Set and Replacement. For SC^M bound both coordinates. Choice in M gives the well-orderings.

(ii) Inside M , take the field generated by the cover from (i) and its relative real algebraic closure in $\mathbf{No}^M$; there are set many polynomials and finitely many roots of each. Quantifier elimination gives elementarity. An all-plus ordinal above every element of E is not in E , hence transcendental over E (a real closed subfield is relatively algebraically closed in an ordered extension), hence over $E[i]$. $\square$

The lemma is not a covering property of a forcing notion, and it does not assert $A \in M$. It holds even when a forcing badly fails familiar small-set covering properties, and the containing ground set can be vastly larger than A . It works because a set of birthdays is bounded and the two universes have the same ordinals.

Lemma 3.3 (Old bounds for new sets; source 03). *Every N -set of elements of $\mathbf{No}^N$ has an upper bound that is an old ordinal. Consequently every N -set of old numbers is bounded above and below in $\mathbf{No}^M$, and $\mathbf{No}^M$ has no set-sized cofinal or coinital subset in N . Old positive elements are coinital in $\mathbf{No}_{>0}^N$.*

Proof. For a set $X \subseteq \mathbf{No}^N$, Replacement bounds all birthdays by an ordinal θ . The all-plus ordinal $\theta + 1$ exceeds every member of X : at any earlier minus sign it wins the first comparison, and it exceeds an all-plus sequence of smaller length. It is old. Apply the same argument to $-X$. For $r \in \mathbf{No}_{>0}^N$ choose an old positive ordinal $\gamma > 1/r$; then $0 < 1/\gamma < r$ and $1/\gamma \in \mathbf{No}^M$. $\square$

Lemma 3.4 (Coding short families; sources 01, 03, 04). *Assume $\text{NS}_{<\kappa}(M, N)$. Every N -set of old surreal or surcomplex numbers of cardinality less than κ is a member of M . More generally, every sequence of length less than κ with values in a fixed ground set is a ground sequence, and every N -subset of cardinality less than κ of a ground set belongs to M .*

Proof. For a ground set B fix in M an injection $e : B \rightarrow \theta$ into an ordinal. Composing a short sequence in B with e gives an ordinal-valued sequence of the same length, hence an old one, and decoding with e shows that the original sequence is old. For an external set A of size less than κ , use lemma 3.2 and an enumeration of A in N of length less than κ ; the enumeration is old, and so is its range. $\square$

The following asymmetric fact is the efficient direction of the main theorem.

Theorem 3.5 (One old side suffices; source 01). *Let L, R be N -sets contained in $\mathbf{No}^M$ with $L < R$. If either $L \in M$ or $R \in M$, then some $x \in \mathbf{No}^M$ satisfies $L < x < R$.*

Proof. Suppose $L \in M$. Choose a ground set $B \supseteq R$ by lemma 3.2 and define, in M , $U = \{b \in B : (\forall \ell \in L) \ell < b\}$. Order is absolute, so $R \subseteq U$ and $L < U$. The ground cut $x = \{L \mid U\}$ satisfies $L < x < U$, and therefore $L < x < R$. This also covers empty sides. If $R \in M$, apply the proved case to $-R < -L$ and negate the separator. $\square$

Corollary 3.6 (Source 01). *If $\text{NS}_{<\kappa}(M, N)$ holds, every external set-sized cut with at least one side of size less than κ is filled in $\mathbf{No}^M$.*

Proof. The small side belongs to M by lemma 3.4; apply theorem 3.5. $\square$

In particular, every N -set of positive elements of $\mathbf{No}^M$ has a positive lower bound in $\mathbf{No}^M$: take $L = \{0\}$. Every N -set of old numbers has an old strict upper bound. These statements survive every extension under consideration, including ones that destroy countable saturation.

4 The first new sequence length

Definition 4.1. When $M \neq N$, the *first new length* is

$$\delta(M, N) = \min\{\lambda : \exists \theta (\lambda^\theta)^N \not\subseteq M\}. \quad (4.1)$$

No value is assigned when $M = N$. A sequence $f : \lambda \rightarrow \text{Ord}$ in N is *fresh over M* (source 04) if $f \notin M$ but $f \restriction \xi \in M$ for every $\xi < \lambda$. This is the forcing-theoretic notion of a fresh function [14]; the name is not claimed.

The minimum exists. *First route (sources 01, 03).* Every set of N can be coded in N by a well-founded extensional relation on an ordinal together with a marked point, and such a relation is coded by a set of ordinals. If all sets of ordinals were old, the code and its Mostowski collapse would be old, so $M = N$. *Second route (sources 04, 08).* Choose a new set x of least rank α . Every set of rank below α is old, so $x \subseteq V_\alpha^M$, and an M -well-ordering of V_α^M identifies x with a new subset of an ordinal. Either way, a new set of ordinals gives a new characteristic sequence.

No finite sequence of old ordinals is new, and a sequence of successor length is determined by its shorter prefix and one old ordinal. Hence $\delta(M, N)$ is an infinite limit ordinal, and every new sequence of length δ is fresh. More generally, a fresh sequence has infinite limit length.

Proposition 4.2 (Regularity of the first new length; sources 01, 03, 04, 08). *The ordinal $\delta(M, N)$ is an infinite regular cardinal in both M and N .*

Proof. Write $\delta = \delta(M, N)$ and choose a new $f : \delta \rightarrow \theta$.

If δ were not an N -cardinal, choose in N a bijection $e : \mu \rightarrow \delta$ with $\mu < \delta$. By minimality $e \in M$, and $f \circ e$, also of length $\mu < \delta$, belongs to M . Decoding with e gives $f \in M$, a contradiction.

If $\text{cf}^N(\delta) = \mu < \delta$, choose a cofinal $e : \mu \rightarrow \delta$; it is old by minimality. Every $f \restriction e(\xi)$ is old and lies in the ground set $T = ({}^{<\delta}\theta)^M$. Well-order T in M . The sequence of codes of $f \restriction e(\xi)$, $\xi < \mu$, has length less than δ and is old; its ground decoding and union reconstruct f , a contradiction. Hence δ is regular in N . A ground bijection or a shorter ground cofinal sequence would survive to N , so δ is a regular cardinal of M as well. $\square$

Sources 01, 03 and 08 state regularity in both universes; source 04 states the N clause, and its proof gives the M clause by the last sentence above. This is an elementary minimal-length fact about extensions, not a new characterization of regular cardinals and not a new forcing invariant.

Remark 4.3 (Ordinal sequences, not binary ones; source 08). Ordinal-valued sequences in (4.1) must not be replaced by subsets of small ordinals. A new countable cofinal sequence in a large cardinal may exist without any new real; Prikry forcing is the decisive example (section 13.4). The codings below preserve the number of bounds rather than encode large ordinal entries into a short binary string.

5 Coding a fresh sequence by an omitted cut

The converse direction of the main theorem needs more than the observation that a new binary sign has a new cut. A short sequence can take values in a very large ordinal: a new countable cofinal sequence can appear while no real is added. Encoding it as one binary sign might require a long ordinal domain, which would destroy the cardinal bound on the cut. Each construction below keeps the number of inequalities equal to the length of the sequence, independently of the size of its values. Section 5.1 is the main route (source 01); sections 5.2 and 5.3 are marked second routes (sources 04 and 08, with source 03's variant in proposition 9.1).

5.1 Main route: an arbitrarily branching interval tree (source 01)

Lemma 5.1 (Arbitrarily branching interval tree). *Let $\theta \geq 2$ be an ordinal and λ a nonzero limit ordinal. In M there is a set-sized table of pairs*

$$(a_s, b_s) \in (\mathbf{No}^M)^2, \quad s \in T := ({}^{<\lambda}\theta)^M,$$

such that:

- (i) $a_\emptyset = 0 < b_\emptyset = 1$;
- (ii) if s is a proper initial segment of t , then $a_s < a_t < b_t < b_s$;
- (iii) the intervals of distinct immediate successors of one node are pairwise disjoint and ordered by their successor labels.

Proof. Recurse on the levels below λ . Given (a_s, b_s) put $w_s = b_s - a_s > 0$, regard ξ and θ as their all-plus surreal images, and set

$$a_{s \smallfrown \xi} = a_s + w_s \frac{3\xi + 1}{3\theta + 3}, \quad b_{s \smallfrown \xi} = a_s + w_s \frac{3\xi + 2}{3\theta + 3} \quad (\xi < \theta). \quad (5.1)$$

Every operation here is surreal field arithmetic. If $\xi < \eta < \theta$, the equality of ordinal and surreal successor gives $\xi + 1 \leq \eta$, hence $0 < 3\xi + 1 < 3\xi + 2 < 3\eta + 1 < 3\eta + 2 < 3\theta + 3$. This proves the interior and sibling-separation properties.

At a nonzero limit level $\gamma < \lambda$ and $s \in ({}^\gamma\theta)^M$, the induction hypothesis gives separated ground sets $A_s = \{a_{s \smallfrown \epsilon} : \epsilon < \gamma\}$ and $B_s = \{b_{s \smallfrown \epsilon} : \epsilon < \gamma\}$. Define inside M

$$a_s = \{A_s \mid B_s\}, \quad b_s = \{A_s \cup \{a_s\} \mid B_s\}. \quad (5.2)$$

Then $A_s < a_s < b_s < B_s$. Each level and its prefix data are sets in M ; Replacement and set-length recursion produce the table in M . No recursion along all ordinals is used. $\square$

Theorem 5.2 (Branch decoder and omitted cut). *Suppose $f : \lambda \rightarrow \theta$ belongs to $N \setminus M$, where λ is a nonzero limit ordinal and every proper initial segment $f \upharpoonright \alpha$ belongs to M . Then $\mathbf{No}^M$ omits the finitely satisfiable partial type*

$$p_f(X) = \{ a_{f \upharpoonright \alpha} < X < b_{f \upharpoonright \alpha} : \alpha < \lambda \}. \quad (5.3)$$

It uses at most $|\lambda|$ parameters on each side; the left endpoints strictly increase and the right endpoints strictly decrease.

Proof. All endpoints are old. A finite subset of the inequalities is satisfied by the midpoint of its deepest interval, which is old. Strict nesting proves monotonicity.

Suppose an old x realizes all inequalities. Using only the ground table and x , define $\text{dec}_x : \lambda \rightarrow \theta$ by recursion in M : at a successor step, with prefix s constructed, choose the unique $\xi < \theta$ with $a_{s \smallfrown \xi} < x < b_{s \smallfrown \xi}$ if it exists, and 0 otherwise; at limits take unions. Sibling intervals are disjoint, so the choice is unique, and the default value makes this a *total ground recursion* that does not presuppose a branch in M .

In N , induction gives $\text{dec}_x \upharpoonright \alpha = f \upharpoonright \alpha$ for every $\alpha < \lambda$: at the next step x lies in the interval of $f \upharpoonright (\alpha + 1)$ by the type, and $\alpha + 1 < \lambda$ because λ is limit. Thus $\text{dec}_x = f$, contradicting $\text{dec}_x \in M$ and $f \notin M$. $\square$

The proof does not infer that an external branch is old because all its nodes are old; that inference would be false. An assumed *old realizer* supplies a total, deterministic ground decoder for the whole branch. There is also no assertion that the interval widths tend to zero in the fine topology; section 14 shows they have a common positive old lower bound.

5.2 Second route A: an absolute all-ordinal interval code (source 04)

The tree of lemma 5.1 fixes an alphabet bound θ and is built level by level inside M . Source 04's code instead lets the children of a node be indexed by *all* ordinals and defines the interval of each individual sequence s by set recursion along $\text{dom}(s) + 1$, uniformly in whichever universe contains s . At the root $a_\emptyset = 0$, $b_\emptyset = 1$. With $w_s = b_s - a_s$ and α an ordinal,

$$a_{s \smallfrown \langle \alpha \rangle} = a_s + \frac{w_s}{2\widehat{\alpha} + 3}, \quad b_{s \smallfrown \langle \alpha \rangle} = a_s + \frac{w_s}{2\widehat{\alpha} + 2}, \quad (5.4)$$

all operations being surreal field operations. At a nonzero limit length $\gamma = \text{dom}(s)$,

$$u_s = \{ a_{s \upharpoonright \xi} : \xi < \gamma \mid b_{s \upharpoonright \xi} : \xi < \gamma \}, \quad e_s = \{ 0 \mid u_s - a_{s \upharpoonright \xi}, b_{s \upharpoonright \xi} - u_s : \xi < \gamma \}, \quad (5.5)$$

and $a_s = u_s - e_s/2$, $b_s = u_s + e_s/2$; the right option set of e_s is the union of the two displayed families. Write $I_s = (a_s, b_s)$.

Theorem 5.3 (Absolute ordinal-branching interval code; source 04). (i) *For every proper extension t of s , $[a_t, b_t] \subset (a_s, b_s)$.*

(ii) *If $\alpha < \beta$ then $b_{s \smallfrown \langle \beta \rangle} < a_{s \smallfrown \langle \alpha \rangle}$; distinct children have disjoint closed intervals.*

(iii) *Every nonroot closed interval lies inside $(0, 1)$.*

(iv) *If $s \in M$, both endpoints belong to M and have the same values when computed in M or in N .*

For a fixed input s the construction needs only set recursion along $\text{dom}(s) + 1$; it forms no set of all nodes at a level and no class-valued transfinite recursion.

Proof. At a successor node both denominators in (5.4) exceed one, with $2\widehat{\alpha} + 3 > 2\widehat{\alpha} + 2 > 0$, so $a_s < a_{s \smallfrown \langle \alpha \rangle} < b_{s \smallfrown \langle \alpha \rangle} < b_s$. If $\alpha < \beta$ then $\widehat{\beta} \geq \widehat{\alpha} + 1$, hence $\frac{1}{2\widehat{\beta}+2} \leq \frac{1}{2\widehat{\alpha}+4} < \frac{1}{2\widehat{\alpha}+3}$, which is (ii). At a limit stage the earlier closed intervals are strictly nested, so the cut u_s is legitimate and lies strictly inside every earlier interval; all right options of e_s are positive, so $0 < e_s$ is smaller than each of those distances, and $a_{s \upharpoonright \xi} < u_s - e_s/2 < u_s + e_s/2 < b_{s \upharpoonright \xi}$. This proves (i)–(iii) by induction. For (iv), run the recursion on the prefixes of the given s : successor steps use absolute arithmetic (proposition 3.1(iii)) and limit steps use absoluteness of old cuts (proposition 3.1(ii)) twice; all option sets belong to M by Replacement in M . The values on the prefixes of one input form a set-length sequence of sets, constructed by the ordinary recursion theorem. $\square$

Theorem 5.4 (Fresh-branch omitted cut; source 04). *Let $f : \mu \rightarrow \text{Ord}$ be fresh over M , and put $A_f = \{a_{f \upharpoonright \xi} : \xi < \mu\}$, $B_f = \{b_{f \upharpoonright \xi} : \xi < \mu\}$. Then A_f, B_f are N -sets of old surreals of cardinality at most $|\mu|^N$, $A_f < B_f$, and their strict cut is finitely satisfiable in $\mathbf{No}^M$ but has no separator in $\mathbf{No}^M$. Removing the root endpoints leaves an omitted cut inside $(0, 1)$.*

Proof. All endpoints are old by theorem 5.3(iv), and nesting gives $A_f < B_f$; the midpoint of a deep enough $I_{f \upharpoonright \eta}$ satisfies finitely many constraints. Suppose $x \in \mathbf{No}^M$ separates the whole cut, so $x \in I_{f \upharpoonright \xi}$ for all $\xi < \mu$. Define $g : \mu \rightarrow \text{Ord}$ by set recursion in M : given $g \upharpoonright \xi$, let $g(\xi)$ be the unique ordinal α with $x \in I_{(g \upharpoonright \xi) \smallfrown \langle \alpha \rangle}$ if one exists, and 0 otherwise. The existential quantifier ranges over ordinals, not over a set of all children, and uniqueness comes from (ii). By induction in N , $g(\xi) = f(\xi)$: since μ is a limit, $x \in I_{f \upharpoonright (\xi+1)} = I_{(g \upharpoonright \xi) \smallfrown \langle f(\xi) \rangle}$, an interval computed absolutely from old data. Hence $g = f \in M$, a contradiction. For the last clause omit the index 0 on both sides. $\square$

Proposition 5.5 (Closed formulas along a finite branch; source 04). *For an ordinal α put $r_\alpha = \frac{1}{2\widehat{\alpha}+3}$ and $\ell_\alpha = \frac{1}{(2\widehat{\alpha}+2)(2\widehat{\alpha}+3)}$. If $s = \langle \alpha_0, \dots, \alpha_{n-1} \rangle$, then*

$$a_s = \sum_{j < n} r_{\alpha_j} \prod_{k < j} \ell_{\alpha_k}, \quad b_s = a_s + \prod_{k < n} \ell_{\alpha_k},$$

with the empty product one and the empty sum zero.

Proof. The child interval is the image of $(0, 1)$ under $t \mapsto a_s + w_s(r_\alpha + \ell_\alpha t)$, since $\frac{1}{2\widehat{\alpha}+2} - \frac{1}{2\widehat{\alpha}+3} = \ell_\alpha$. Induct on n . $\square$

For a fresh sequence of length ω , such as a Prikry sequence (section 13.4), all endpoints of the omitted cut are given by these finite rational expressions in the embedded ordinals; no limit-stage cut is needed to write them down. The absence of an old separator is still an infinite assertion, proved by the decoder, not by finite experimentation.

5.3 Second route B: ordinal block codes of signs (source 08)

Here the fresh sequence is written into a single sign sequence whose prefixes provide the bounds.

Lemma 5.6 (Two bounds force a prefix; source 08). *For every sign sequence u , the numerical interval $(u \smallfrown -, u)$ equals $\text{Cone}(u \smallfrown - \smallfrown +)$.*

Proof. An extension of $u \smallfrown - \smallfrown +$ is below u , since its first sign after u is minus, and above $u \smallfrown -$, since its first sign after that is plus. Conversely, if $u \smallfrown - < y < u$ and y first differed from or terminated before the end of u , its comparisons with u and with $u \smallfrown -$ would agree, contradicting the two inequalities. So $u \preceq y$; the

upper inequality forces the next sign to be minus, and the lower inequality, excluding $y = u \frown -$, forces the following sign to be plus. $\square$

Let λ be an infinite limit ordinal and $f : \lambda \rightarrow \text{Ord}$. By set recursion put $p_0 = \emptyset$, $p_{\xi+1} = p_\xi \frown +^{f(\xi)} \frown - \frown +$, and $p_\eta = \bigcup_{\xi < \eta} p_\xi$ at limits, where $+^\alpha$ is a run of plus signs of ordinal length α . Set

$$\text{code}(f) = p_\lambda, \quad u_\xi = p_\xi \frown +^{f(\xi)}, \quad l_\xi = u_\xi \frown -. \quad (5.6)$$

Lemma 5.7 (Decoding and nesting; source 08). *code(f) determines f and its block boundaries by a set-recursive decoder. For $\xi < \eta < \lambda$, $l_\xi < l_\eta < \text{code}(f) < u_\eta < u_\xi$, and $\bigcap_{\xi < \lambda} (l_\xi, u_\xi) = \text{Cone}(\text{code}(f))$. These assertions are absolute for transitive universes containing f.*

Proof. From a block boundary, read the run of plus signs up to the next minus; its order type is $f(\xi)$. Consume that minus and the following plus; the next block begins there, and at limits take suprema of boundaries. By lemma 5.6, $(l_\xi, u_\xi) = \text{Cone}(p_{\xi+1})$. Every later endpoint and $\text{code}(f)$ extend $p_{\xi+1}$; comparing at the final plus marker of block η gives $l_\eta < \text{code}(f)$, and at its minus marker $\text{code}(f) < u_\eta$. A sequence extends every $p_{\xi+1}$ iff it extends their union. $\square$

Theorem 5.8 (Fresh-sequence cut and its separators; source 08). *Let $f : \lambda \rightarrow \text{Ord}$ be fresh over M , λ an infinite limit ordinal. The endpoints l_ξ, u_ξ are old, but the cut $L_f = \{l_\xi : \xi < \lambda\}$, $R_f = \{u_\xi : \xi < \lambda\}$ has no separator in $\mathbf{No}^M$. In $\mathbf{No}^N$ its class of separators is exactly $\text{Cone}(\text{code}(f))$. Every proper prefix of $\text{code}(f)$ is old, while $\text{code}(f)$ is new. If $\lambda = \delta(M, N)$, the cut determines a set-presented gap of $\mathbf{No}^M$ of exact cofinality pair (λ, λ) , and $\text{cf}^N \text{b}(\text{code}(f)) = \lambda$. (For an arbitrary fresh sign, the cofinalities of its gap are computed in lemma 8.3.)*

Proof. Since λ is a limit, each $f \upharpoonright (\xi + 1)$ is old, and computing its code inside M gives $p_{\xi+1}, u_\xi, l_\xi$. The two endpoint sets need not belong to M . By lemma 5.7 the separators in $\mathbf{No}^N$ are the extensions of $\text{code}(f)$. An old separator y would give $\text{code}(f) \preceq y$, hence $\text{code}(f) \in M$ by initiality, and decoding inside M would recover f . The same decoding shows $\text{code}(f) \notin M$. Any proper prefix of $\text{code}(f)$ is a restriction of some old p_α .

For the gap, recall that $\lambda = \delta$ is a regular cardinal of N (proposition 4.2). The classes $D^- = \{y \in \mathbf{No}^M : \exists \xi \ y \leq l_\xi\}$ and $D^+ = \{y \in \mathbf{No}^M : \exists \xi \ y \geq u_\xi\}$ partition $\mathbf{No}^M$, since a point in neither would separate the cut; strict nesting shows that neither has an endpoint toward the other. A cofinal $X \subseteq D^-$ of size below λ is impossible: its indices ξ_x with $x \leq l_{\xi_x}$ are bounded below λ by regularity, and a later l_η exceeds X ; symmetrically on the right. Finally $\text{b}(\text{code}(f)) = \sum_{\xi < \lambda} (f(\xi) + 2)$; its increasing partial sums are cofinal, and fewer than λ cofinal ordinals would, by regularity, bound the partial-sum indices below λ . $\square$

The distinction between a family of old endpoints and an old *set* of endpoints is the whole mechanism: internal saturation fills cuts whose bounding sets are in M , and the cut constructed here is an external set of old bounds. Each possibly enormous value $f(\xi)$ is stored in a single old sign prefix; what is compressed is the *number of parameters*, not the birthdays of their values. Source 03 uses the variant $\text{code}^+(f) = \frown_{\xi} (+^{f(\xi)+1} \frown -)$, with endpoints read off before the last plus and before the minus of each block; it is printed with its proof in proposition 9.1, where it serves the least-gap theorem. Both codes use ordinal sums, not the natural sums that describe addition of ordinal surreals.

6 The exact spectrum and its first gap

6.1 Proof of the main equivalence

Proof of theorem 1.1. (i) $\Rightarrow$ (ii) is corollary 3.6, and (ii) $\Rightarrow$ (iii) is immediate. Suppose (i) fails. Choose the least ordinal $\lambda < \kappa$ for which there is a new ordinal-valued sequence of length λ , choose one such sequence, and bound its range by an ordinal $\theta \geq 2$. As observed in section 4, λ is an infinite limit ordinal and the sequence is fresh. Theorem 5.2 produces an unfilled cut whose two sides together have cardinality at most $|\lambda| < \kappa$ in N , so (iii) fails; theorem 5.4 gives one inside $(0, 1)$. For uncountable κ , proposition 3.1 and lemma 2.3(i) identify (iii) with ordered-field κ -saturation, and lemma 2.3(ii) with κ -saturation of the dense order. $\square$

κ need not be regular. Only the conversion to real closed saturation requires $\kappa > \omega$; at $\kappa = \omega$ both sides of (i) $\Leftrightarrow$ (iii) hold trivially (source 08), since finite sequences of ordinals and finite sets of old numbers are old. Source 03 states the cut equivalence for every infinite *ordinal* bound:

$$\mathbf{No}^M \text{ has } \eta_\kappa \iff \forall \alpha < \kappa \left({}^\alpha\text{Ord} \right)^N \subseteq M, \quad (6.1)$$

where the right side is shorthand for the class statement that every ordinal-valued sequence of length α in N belongs to M . The proof is the same: a new sequence of length below κ gives $\delta < \kappa$ and an omitted pair of size δ (theorem 6.2).

6.2 Exact two-sided cofinalities

Definition 6.1 (Set-presented gaps). A *set-presented gap* of $\mathbf{No}^M$ is a partition $\mathbf{No}^M = D \sqcup U$ with $D < U$, neither D having a greatest nor U a least element, together with N -sets $A \subseteq D$ cofinal in D and $B \subseteq U$ coinital in U . Under (H) a gap is given by such a presentation (A, B) , and $D = \{x \in \mathbf{No}^M : \exists a \in A \ x \leq a\}$, $U = \mathbf{No}^M \setminus D$ are defined from it; statements about all gaps are schemes over presentations. Its *cofinality pair* ($\text{cf}^N D, \text{ci}^N U$) is the pair of least sizes, computed in N , of cofinal and coinital N -subsets; these are infinite regular cardinals. D and U themselves are never required to be sets. This is source 01's "set-character gap", source 03's "set-presented gap" and source 08's "Dedekind gap" with its character.

Theorem 6.2 (First gap theorem; sources 01, 03, 08). *Let $M \neq N$ and $\delta = \delta(M, N)$. Every set-presented gap of $\mathbf{No}^M$ has both cofinalities at least δ , and $\mathbf{No}^M$ has one of exact cofinality pair (δ, δ) .*

Proof. Apply theorem 5.2 to a new sequence of minimal length δ and write its endpoints as a_α, b_α . Define in N $D = \{x \in \mathbf{No}^M : \exists \alpha < \delta \ (x \leq a_\alpha)\}$ and $U = \{x \in \mathbf{No}^M : \exists \alpha < \delta \ (b_\alpha \leq x)\}$. A point in neither class would realize the omitted type, so $D \cup U = \mathbf{No}^M$ and $D < U$; strict monotonicity and the limit length show that neither class has an endpoint toward the other. The endpoint sequences are witnesses of size δ .

For the lower bound, take any set-presented gap with witnesses A and B . If either could have size less than δ , theorem 1.1(ii) at $\kappa = \delta$ would give an old strict separator between A and B . A point of the lower class cannot lie strictly above a cofinal subset of it, and a point of the upper class cannot lie strictly below a coinital subset of it; since the classes partition $\mathbf{No}^M$, this is impossible. $\square$

The block code gives a second proof of the upper bound, with the exact separator class $\text{Cone}(\text{code}(f))$ (theorem 5.8, source 08).

Corollary 6.3 (Recovering the threshold from the order; sources 01, 03, 08). *For $M \neq N$, $\delta(M, N)$ is the least cardinal μ for which $\mathbf{No}^M$ has an unfilled set-sized cut with both sides of size at most μ ; equivalently, the least cofinality on either side of a set-presented gap of $\mathbf{No}^M$.*

Proof. The upper bounds follow from theorem 6.2. A smaller unfilled cut contradicts the main theorem at δ , and a smaller gap cofinality contradicts the first gap theorem. $\square$

Corollary 6.4 (The uncountable spectrum; sources 01, 03, 04, 08). *For uncountable N -cardinals κ , $\mathbf{No}^M$ is κ -saturated exactly when $\kappa \leq \delta(M, N)$. Thus if $\delta > \omega$ it is δ -saturated but not $(\delta^+)^N$ -saturated, and if $\delta = \omega$ it is not ω_1 -saturated. $\mathbf{No}^M$ is externally set-saturated if and only if $M = N$ (source 04).*

Proof. There are no new sequences of length less than δ , and there is one of length δ ; apply the main theorem. If $M = N$, use lemma 2.3(iii); if $M \neq N$, any uncountable cardinal above δ fails. $\square$

That $\mathbf{No}^M$ has no smaller set-presented gap does not exclude cuts with proper-class cofinality on one side. Those are outside the invariant and outside the main interpolation theorem. The definition of the invariant uses only the order of $\mathbf{No}^M$, external sets and their cardinalities, yet its value is the first length at which N has new ordinal data. Since a real closed field has a unique field order, it is preserved even by isomorphisms stated only in the field language (source 03).

7 The countable boundary: no new reals is weaker

The absence of new reals and the absence of new countable ordinal-valued sequences are different preservation properties, and the old surreal field records the difference exactly.

Theorem 7.1 (Finite-parameter saturation; sources 01, 03, 04, 08).

$$\mathbf{No}^M \text{ is } \omega\text{-saturated in } N \iff \mathcal{P}(\omega)^M = \mathcal{P}(\omega)^N \iff \mathbb{R}^M = \mathbb{R}^N.$$

In contrast, the pure order $(\mathbf{No}^M, <)$ is always ω -saturated (source 08).

Proof. Equality of the power sets is equivalent to equality of the reals by ordinary countable coding in both transitive universes.

Suppose no new subsets of ω are added. Let A be a finite set of old parameters and p a complete one-type over A finitely satisfiable in $\mathbf{No}^M$ according to N . The finite set A is in M , the formulas over A have a countable enumeration in M , and the code of p is a subset of ω , so $p \in M$. For each finite $p_0 \subseteq p$, solvability of its conjunction is an existential ordered-field formula over old parameters, with the same truth value in M and N by proposition 3.1 and quantifier elimination. Hence M recognizes p as finitely satisfiable, and in M the full surreal field realizes every set-sized type (lemma 2.3(iii)).

Conversely let $r \in \mathbb{R}^N \setminus \mathbb{R}^M$ and consider the rational-cut partial type

$$\{q < X : q \in \mathbb{Q}, q < r\} \cup \{X < q : q \in \mathbb{Q}, r < q\}. \quad (7.1)$$

It is finitely satisfiable by rationals, and it has no parameters, since $m/n < X$ is equivalent to $m < nX$ for $n > 0$. If an old x realized it, the ground set $\{q \in \mathbb{Q} : q < x\}$ would be the lower rational cut of r , and ground Dedekind completeness would put r in M . For a complete omitted type, take the type of the image of r in $\mathbf{No}^N$ over the empty set; every finite part is realized in $\mathbf{No}^M$ because $\mathbf{No}^M \preccurlyeq \mathbf{No}^N$. No class compactness principle is required. Source 04 gives the same argument with the dyadic prefixes q_n of a new sign $s : \omega \rightarrow \{-, +\}$, using the parameter-free formulas $q_n < X$ for $s(n) = +$ and $X < q_n$ for $s(n) = -$.

Finally, all finite cuts of the old order are filled, so lemma 2.3(ii) gives ω -saturation of the pure order. $\square$

Corollary 7.2 (Source 01). *An extension that adds no reals but adds a countable ordinal-valued sequence leaves $\mathbf{No}^M$ ω -saturated and destroys ω_1^N -saturation. It creates an (ω, ω) set-presented gap in $\mathbf{No}^M$, but none of its omitted types can be witnessed over a fixed finite parameter set.*

Proof. The first new length is ω . Apply theorems 6.2 and 7.1 and corollary 6.4. The last assertion is the surviving ω -saturation. $\square$

A new countable ordinal-valued sequence need not code a new real. The new countable parameter family in the interval tree carries information that no old finite parameter tuple can code; a bare new-real argument would miss this example. One must not set $\kappa = \omega$ in the field-saturation clause of theorem 1.1: the real closure of a finite parameter set is usually countable, not finite. This is not a conflict between two definitions of saturation but an exact distinction between finite and countable parameter sets in the same language.

Part II

Fresh signs classify the set-presented gaps (source 03)

8 The gap–fresh-sign correspondence

This part is source 03's. It classifies *all* set-presented gaps of $\mathbf{No}^M$, not only the first. It answers source 01's question about asymmetric cofinality pairs (section 19). Its proofs use only the sign order, prefix-initiality, the old constant-tail closure and the set-cut principle in N ; they need no Global Choice.

Definition 8.1 (Fresh signs). *A fresh sign over M in N is $s \in \mathbf{No}^N \setminus \mathbf{No}^M$ such that $s \upharpoonright \alpha \in \mathbf{No}^M$ for every $\alpha < b(s)$. Write $\text{Fresh}(M, N)$ for the class of fresh signs. This is the binary special case of a fresh function (definition 4.1); the forcing notion of freshness and its spectra are due to Fischer–Koelbing–Wohofsky [14], and the name is not claimed as a new invariant. For $s \in \mathbf{No}^N \setminus \mathbf{No}^M$ put*

$$D_s = \{x \in \mathbf{No}^M : x < s\}, \quad U_s = \{x \in \mathbf{No}^M : s < x\}.$$

Write $\text{GapSpec}_N(\mathbf{No}^M)$ for the class of cofinality pairs of set-presented gaps of $\mathbf{No}^M$ (definition 6.1), computed in N .

Lemma 8.2 (Both signs are cofinal). *If $s \in \text{Fresh}(M, N)$ and $\lambda = b(s)$, then λ is a nonzero limit ordinal, and both $P_s = \{\alpha < \lambda : s(\alpha) = +\}$ and $Q_s = \{\alpha < \lambda : s(\alpha) = -\}$ are cofinal in λ .*

Proof. A successor-length sequence is determined by its immediately preceding prefix and one sign; the empty sequence is old. If, say, P_s were bounded by $\alpha < \lambda$, then s would be the old prefix $s \upharpoonright \alpha$ followed by minus signs up to the old ordinal λ , an operation in M . Interchange the signs for Q_s . $\square$

Lemma 8.3 (Canonical cofinal presentations). *For a fresh sign s of length λ put $A_s = \{s \upharpoonright \alpha : \alpha \in P_s\}$ and $B_s = \{s \upharpoonright \alpha : \alpha \in Q_s\}$. Then A_s is cofinal in D_s , B_s is coinital in U_s , both are N -sets of old numbers, $A_s < s < B_s$, and*

$$\text{cf}^N D_s = \text{ci}^N U_s = \text{cf}^N \lambda. \quad (8.1)$$

Proof. At a plus position the prefix ends before a plus, so lies below s ; at a minus position above. These prefixes are old by freshness, and no old element extends s , since restriction to λ would place s in M .

Take $x \in D_s$. Either x is a proper prefix of s or it first disagrees with s at some $\gamma < \lambda$. Choose a plus position beyond γ (or beyond $b(x)$); comparison at the earlier position gives $x < s \upharpoonright \alpha < s$. A late minus position handles U_s . This also shows that neither side has an endpoint.

Write $\mu = \text{cf}^N \lambda$. A cofinal sequence of positions of length μ can be replaced by later plus and later minus positions, giving witnesses of size at most μ . Conversely, if $X \subseteq D_s$ has size below μ , record for each $x \in X$ a position where x ends as a prefix or first disagrees with s ; these positions are bounded below λ , and a plus prefix beyond the bound exceeds all of X and lies in D_s . The right side is symmetric. $\square$

Theorem 8.4 (Gap–fresh-sign correspondence). *The assignment $s \mapsto (D_s, U_s)$ is a bijection between fresh signs over M in N and the set-presented gaps of $\mathbf{No}^M$. The inverse sends a gap with any presentation $A \subseteq D$, $B \subseteq U$ to the simplest separator $\{A \mid B\}$ computed in N , and the result does not depend on the chosen presentation. Every set-presented gap is symmetric, with cofinalities given by (8.1).*

Proof. Lemma 8.3 shows the assignment is well defined. For surjectivity, let (D, U) be presented by A, B . No member of $\mathbf{No}^M$ separates A, B : a member of D is at most some member of A , and a member of U is at least some member of B . Let $s = \{A \mid B\}^N$; then $s \notin \mathbf{No}^M$. Every proper prefix $y = s \upharpoonright \gamma$ is old: by simplicity y does not separate A, B ; if $y < s$, all inequalities $y < b$ already hold, so there is $a \in A$ with $y \leq a < s$, and convexity of $\text{Cone}(y) \ni s$ gives $y \preceq a$, so y is a prefix of an old number; if $s < y$, argue with some $b \in B$. Thus s is fresh. Cofinality of A gives $D \subseteq D_s$ and coinitality of B gives $U \subseteq U_s$; both are partitions of $\mathbf{No}^M$, so the gap is (D_s, U_s) .

For injectivity, take distinct fresh s, t . Neither is a proper prefix of the other, since then a new sign would be a proper prefix of a fresh sign. Their common prefix u is old and lies strictly between them, on opposite sides of the two cuts. The same argument shows independence of the presentation. Symmetry is lemma 8.3. $\square$

Theorem 8.5 (The whole filling class). *If s is fresh, then $\{y \in \mathbf{No}^N : D_s < y < U_s\} = \text{Cone}(s)$. Every new $x \in \mathbf{No}^N \setminus \mathbf{No}^M$ lies in exactly one such cone, that of its shortest new prefix. Thus $\mathbf{No}^N \setminus \mathbf{No}^M$ is the disjoint union of the cones $\text{Cone}(s)$, $s \in \text{Fresh}(M, N)$, as class-defined convex pieces.*

Proof. An old element cannot extend s , so its comparison with any extension of s is decided before $b(s)$ and equals its comparison with s ; this gives $\supseteq$. If y does not extend s , either it is a proper prefix of s , hence old and not a filler of its own ground cut, or it first disagrees before $b(s)$; a later plus prefix in D_s or a later minus prefix in U_s then lies on the wrong side of y . For a new x , the least length of a new prefix gives a fresh prefix, and distinct fresh signs are incomparable, so their cones are disjoint. The union is shorthand for membership in exactly one piece; it posits no set of all pieces. $\square$

Corollary 8.6 (Exact gap spectrum). *With all cofinalities taken in N ,*

$$\text{GapSpec}_N(\mathbf{No}^M) = \{(\text{cf}^N b(s), \text{cf}^N b(s)) : s \in \text{Fresh}(M, N)\}. \quad (8.2)$$

In particular, an asymmetrically set-presented gap cannot arise in a same-ordinal ground surreal field.

This is stronger than the existence of a missing cut after adding a real. It classifies all set-presented gaps, including gaps with long sign codes and small outer cofinality, and it explains why two new surreals can have exactly the same cut over the whole old field: they lie in one fresh cone, which can contain a proper class of extensions. The fresh signs themselves, not only their lengths, parametrize the gaps. The symmetry concerns the two sides of a bounded gap, not the global cofinality of the field, which is a proper class (lemma 3.3). Cofinalities must be computed in N : the old universe's cofinality of $b(s)$ can be the wrong one (section 13.4).

9 The least gap

Proposition 9.1 (Terminal-plus block code; source 03). *Let μ be an infinite regular cardinal of N , and let $f : \mu \rightarrow \text{Ord}$ be new with $f \upharpoonright \xi \in M$ for every $\xi < \mu$. Define, by ordinal concatenation,*

$$\text{code}^+(f) = \bigcap_{\xi < \mu} (+^{f(\xi)+1} \frown -), \quad \text{b}(\text{code}^+(f)) = \sum_{\xi < \mu} (f(\xi) + 2).$$

Then $\text{code}^+(f)$ is fresh over M and $\text{cf}^N \text{b}(\text{code}^+(f)) = \mu$. With $\sigma_\xi = \sum_{\eta < \xi} (f(\eta) + 2)$, the old prefixes

$$a_\xi = \text{code}^+(f) \upharpoonright (\sigma_\xi + f(\xi)), \quad b_\xi = \text{code}^+(f) \upharpoonright (\sigma_\xi + f(\xi) + 1)$$

satisfy $a_\xi < a_\eta < \text{code}^+(f) < b_\eta < b_\xi$ for $\xi < \eta < \mu$, and present an omitted (μ, μ) gap of $\mathbf{No}^M$.

Proof. Each block is a nonempty plus run of successor length followed by one minus. A proper prefix of the concatenation lies inside some block and is determined by the old restriction $f \upharpoonright (\xi + 1)$ and ordinal positions; since μ is a limit, $\xi + 1 < \mu$, so the prefix is old. If $\text{code}^+(f)$ were in M , its minus positions would delimit the blocks, and the unique predecessors of the successor run lengths would recover f in M . So it is fresh.

The partial sums σ_ξ increase and are cofinal, so $\text{cf}^N \text{b}(\text{code}^+(f)) \leq \mu$; fewer than μ cofinal positions would, by regularity, bound the block indices below μ . At length $\sigma_\xi + f(\xi)$ the next sign is the last plus of block ξ , and at $\sigma_\xi + f(\xi) + 1$ it is the minus; first-disagreement comparison gives the nesting. The endpoint lengths are cofinal, so, as in lemma 8.3, the endpoints are cofinal and coinital in the two sides, and theorem 8.4 gives an omitted gap with the asserted cofinalities. $\square$

The extra plus in each run is not cosmetic: for a limit value $f(\xi)$, a run of length $f(\xi)$ has no last plus, and the uniformly specified left endpoint would not exist. Source 08's code in section 5.3 achieves the same with the two terminal signs $- \frown +$.

Theorem 9.2 (Exact least-gap theorem; source 03). *For $M \neq N$,*

$$\delta(M, N) = \min\{|A \cup B|^N : A < B \text{ is an omitted set pair in } \mathbf{No}^M\} \quad (9.1)$$

$$= \min\{\text{cf}^N \text{b}(s) : s \in \text{Fresh}(M, N)\}. \quad (9.2)$$

There is an omitted gap of exact cofinality pair (δ, δ) .

Proof. The lower bound in (9.1) is the main theorem at $\kappa = \delta$ (equivalently lemma 3.4: a pair of size below δ is old, and its old cut fills it). Apply proposition 9.1 to a new $f : \delta \rightarrow \text{Ord}$, using proposition 4.2; its endpoint families have size δ . Every fresh sign gives an omitted presentation of size $\text{cf}^N \text{b}(s)$ by lemma 8.3 and theorem 8.4, so no such cofinality is below δ , and the block code attains δ . $\square$

The recovery is deliberately limited. The invariant recovers a threshold, and corollary 8.6 recovers cofinality data. It does not recover the inner model, the forcing notion, or the actual sign lengths from a bare ordered-field presentation. Different grounds can have the same threshold, and equality of gap spectra is not proved to classify the fields.

Part III

The first new birthday (source 08)

10 First new birthdays and their relation to saturation

The first new surreal number, measured by birthday, need not appear at the same ordinal as the first omitted-cut size.

Definition 10.1 (First new birthday). For $M \neq N$ put

$$\beta(M, N) = \min\{\alpha : \mathcal{P}(\alpha)^M \neq \mathcal{P}(\alpha)^N\}, \quad (10.1)$$

the power sets being compared as actual sets of subsets in N .

Proposition 10.2 (Birthday threshold; source 08). $\beta(M, N) = \min\{b(s) : s \in \mathbf{No}^N \setminus \mathbf{No}^M\}$. It is an infinite cardinal in both M and N . With $\delta = \delta(M, N)$ and $\beta = \beta(M, N)$,

$$\delta \leq \text{cf}^N(\beta) \leq \beta. \quad (10.2)$$

In particular, the first new birthday can be much larger than the least size of an omitted cut.

Proof. Subsets of α and signs of length α are uniformly interdefinable by characteristic signs. This gives the birthday characterization; existence follows from section 4. No finite subset is new.

If β were not an M -cardinal, an M -bijection between β and a smaller ordinal would pull a new subset of β back to a new subset of that smaller ordinal. Suppose there were an N -bijection $e : \mu \rightarrow \beta$ with $\mu < \beta$. Transporting the order of β along e gives a well-order $\triangleleft$ of μ of type β . Since β is an infinite M -cardinal, M codes $\mu \times \mu$ by an ordinal $\gamma < \beta$, so by minimality $\triangleleft \in M$. Its order type is absolute, so M would see a well-order of μ of type β , contradicting that β is an M -cardinal.

For the first inequality, choose a new $X \subseteq \beta$ and let $\mu = \text{cf}^N(\beta)$. If $\mu < \delta$, a cofinal $c : \mu \rightarrow \beta$ belongs to M , and every $X \cap c(\xi)$ is old by minimality of β . In M well-order $S = \bigcup_{\alpha < \beta} \mathcal{P}(\alpha)^M$ and code these restrictions by ordinals; the code sequence has length $\mu < \delta$, so it is old, and decoding and taking the union in M recovers X . Hence $\delta \leq \text{cf}^N(\beta)$. $\square$

Remark 10.3 (Two resources; source 08). β measures the least *length of a new sign sequence*; δ measures the least *number of old parameters needed for an omitted cut*. Long old parameters can carry large ordinal entries, as the block codes show. Equality of all low birthday fragments therefore does not by itself guarantee countable saturation of the whole old field (corollary 13.4).

Remark 10.4 (β is the least length of a fresh sign; merge observation). A new sign of least birthday β has all its proper prefixes old, since they are signs of smaller birthday; so it is fresh. Every fresh sign is new, so $\beta = \min\{b(s) : s \in \text{Fresh}(M, N)\}$. By theorem 9.2, δ is the least *cofinality* $\text{cf}^N b(s)$ of a fresh sign. This two-line observation, added in the merge, places source 08's β in source 03's fresh-sign picture. It does not answer source 03's question whether an intrinsic ordered-field invariant recovers such lengths (question 19.5): β is defined through birthdays, which are not part of the field language.

Remark 10.5 (Relation to the sibling report). The sibling report docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets proves, for every same-ordinal outer model, that the first new birthday is the least ordinal acquiring a new subset; it also writes that ordinal β . Proposition 10.2 keeps source 08's statement and proof because source 08 adds that β is a cardinal of N and the inequality $\delta \leq \text{cf}^N \beta$

with the first new length δ of this report. The sibling report also studies bounded birthday fields in a language with a birthday symbol; section 12 here concerns the ordered-field saturation of such fragments, without that symbol.

Part IV

Surcomplex fields, fragments, forcing and topology

11 Surcomplex numbers: a language-dependent dichotomy

Let $c_M(a + bi) = a - bi$ on SC^M and c_N likewise on SC^N . The collection's automorphism report already emphasizes the difference between a pure algebraically closed field and a field with a distinguished surreal real axis [19]. That distinction is standard algebra and is not claimed as new. The point here is its exact effect on *external saturation under a change of universe*.

11.1 The pure field forgets the sequence obstruction

Lemma 11.1 (Algebraic closure stays a set; source 03). *Let H be a class field and $A \subseteq H$ an N -set. The field generated by A is a set, and the elements of H algebraic over it form a set. If H is algebraically closed, this relative algebraic closure is an algebraically closed set subfield of H .*

Proof. Finite rational expressions in A form a set, there are set many polynomials over that field, and a nonzero polynomial of degree d has at most d roots. Collect the root sets by Replacement. A root of a polynomial with algebraic coefficients is algebraic over the original field. $\square$

Theorem 11.2 (Set-saturation of the old pure surcomplex field; sources 01, 03, 04, 08). *For every N -set $A \subseteq SC^M$, every complete one-variable pure-field type over A that is finitely satisfiable in SC^M is realized in SC^M . Consequently SC^M is κ -saturated in N for every N -cardinal κ , with no assumption on new sequences or new reals. More generally (source 03), every definable proper-class algebraically closed field is saturated in the pure field language over every N -set of parameters.*

Proof. Choose a ground set $B \supseteq A$ by lemma 3.2. In M let E be the set of elements of SC^M algebraic over $\mathbb{Q}(B)$; it is an algebraically closed ground subfield by lemma 11.1. Choose an old $t \notin E$ without a class-sized transcendence basis: bound the birthdays of the coordinates of all elements of E and take a larger all-plus ordinal. Then t is transcendental over $\mathbb{Q}(B)$, and this persists in N , because a polynomial witnessing algebraicity has a finite old coefficient tuple. In particular t is transcendental over $\mathbb{Q}(A)$ as formed in N .

If the given type p contains no equation $P(X) = 0$ with $P \neq 0$ over $\mathbb{Q}(A)$, completeness and quantifier elimination make p the unique transcendental type over $\mathbb{Q}(A)$, realized by t . Otherwise choose such an equation. Its finitely many roots lie in SC^M by proposition 3.1. If none realized p , choose for each root a formula of p that it fails; together with $P(X) = 0$ these form an unsatisfiable finite subset of p . The argument covers algebraic parameters and any new organization of A .

For a general proper-class algebraically closed field H , let D be the algebraic closure in H of the field generated by A (lemma 11.1); extend the type to D by compactness in that set model, realize an algebraic extension by a root in D , and a transcendental one by any $y \in H \setminus D$, which exists because H is a proper class. $\square$

There is no cardinal paradox: SC^M is a proper class of N , and the statement is a scheme over set-sized parameter sets, not a claim that a set of cardinality μ realizes types over itself of size μ . The theorem is an established model-theoretic mechanism rather than a new principle about algebraically closed fields; what matters is its contrast with the real field. Nonalgebraic one-types over an algebraically closed field have no order cut to remember. Nor does the theorem by itself give a class isomorphism $SC^M \cong SC^N$; that needs a class back-and-forth and Global Choice (part V).

11.2 Naming conjugation restores the old ordered field

Lemma 11.3 (Real coordinates are interpretable; sources 01, 03, 04, 08). *The fixed field of c_M is definable in (SC^M, c_M) by $c_M(x) = x$, and on it the order is definable by*

$$x < y \iff \exists u (c_M(u) = u \wedge u \neq 0 \wedge u^2 = y - x). \quad (11.1)$$

After naming i , the map $z \mapsto (\operatorname{re} z, \operatorname{im} z)$ with

$$\operatorname{re} z = \frac{z + c_M(z)}{2}, \quad \operatorname{im} z = \frac{z - c_M(z)}{2i} \quad (11.2)$$

is a definable bijection $SC^M \rightarrow (\mathbf{No}^M)^2$, and field operations and conjugation become polynomial operations on coordinates: $(a, b)(u, v) = (au - bv, av + bu)$ and $c_M(a, b) = (a, -b)$.

Proof. The fixed-point equation gives zero imaginary coordinate; in a real closed field the nonzero squares are exactly the positive elements. The coordinate formulas follow from $i^2 = -1$, with inverse $(a, b) \mapsto a + bi$. Quantifiers over SC^M become pairs of quantifiers over $\mathbf{No}^M$, and quantifiers over $\mathbf{No}^M$ become quantifiers over the definable fixed field. $\square$

Theorem 11.4 (Conjugation spectrum; sources 01, 03, 04, 08). *For every infinite N -cardinal κ ,*

$$(SC^M, c_M) \text{ is } \kappa\text{-saturated in } N \iff \mathbf{No}^M \text{ is } \kappa\text{-saturated in } N.$$

For uncountable κ this is equivalent to $NS_{<\kappa}(M, N)$, and for $\kappa = \omega$ to the absence of new reals. The statements remain equivalent after naming i . Moreover the inclusion $(SC^M, c_M) \preccurlyeq (SC^N, c_N)$ is elementary (sources 03, 04, 08).

Proof. Naming a fixed finite tuple of constants preserves and reflects κ -saturation for infinite κ : absorb the tuple into the parameter set, or extend a reduct type by compactness and realize it. So we may name i . If $\mathbf{No}^M$ is κ -saturated, translate a type in (SC^M, c_M, i) over fewer than κ parameters by lemma 11.3: the real and imaginary coordinates of the parameters are still fewer than κ , the translated type has two field variables, and finite-variable saturation of $\mathbf{No}^M$ realizes it. Conversely translate a type over a small subset of $\mathbf{No}^M$ into (SC^M, c_M) by (11.1), restricting quantifiers to the fixed field and adding $c_M(X) = X$; saturation gives a fixed-field realization. The last assertions follow from theorems 1.1 and 7.1. Elementarity: the same coordinate translation reduces each formula with old parameters to an ordered-field formula, and $\mathbf{No}^M \preccurlyeq \mathbf{No}^N$. $\square$

In particular the interval type (5.3) remains omitted in the conjugation expansion when written with the definable fixed field and definable positive squares. In the pure field those order inequalities are unavailable, and the one-variable transcendental type is realized by theorem 11.2.

Remark 11.5 (Conjugation is not definable in the pure field; source 08). In a pure algebraically closed field every definable subset of one variable, with any finite tuple of parameters, is finite or cofinite. The fixed field of c_M is infinite with infinite complement, so c_M is not definable in the pure field, even with parameters, and naming a set of constants does not change this. The contrast in table 3 therefore does not violate preservation of saturation under definitional expansions.

Table 3: Saturation in N by structure and language (source 08). “No new reals” means $\mathbb{R}^M = \mathbb{R}^N$.

Structure and language	ω -saturation	Uncountable κ -saturation
$(\mathbf{No}^M, <)$, pure order	Always	$\text{NS}_{<\kappa}(M, N)$
$\mathbf{No}^M$, ordered field	No new reals	$\text{NS}_{<\kappa}(M, N)$
SC^M , pure field	Always	Always
(SC^M, c_M)	No new reals	$\text{NS}_{<\kappa}(M, N)$

11.3 A nonisomorphism consequence

Corollary 11.6 (Sources 01, 04, 08). *Assume $M \neq N$. There is no N -class order isomorphism $\mathbf{No}^M \rightarrow \mathbf{No}^N$ whose restrictions to sets have set-sized images. Consequently there is no such field isomorphism, and no such isomorphism $(\text{SC}^M, c_M) \cong (\text{SC}^N, c_N)$.*

Proof. The first gap theorem supplies two N -sets of old endpoints with no separator in $\mathbf{No}^M$. An isomorphism as stated sends them to separated N -sets in $\mathbf{No}^N$, whose cut has a separator; its inverse image would separate the original cut. A field isomorphism of real closed fields preserves order, since positivity is being a nonzero square, and an isomorphism preserving conjugation restricts to an isomorphism of the definable fixed fields. $\square$

The set-image condition is automatic for definable class maps and for class maps of GBC with class Replacement; sources 04 and 08, working in a class theory, state the corollary without it. It is stated here to avoid quantifying over unrestricted external graphs without an ambient class theory. The corollary compares an inner universe with its ambient universe only; it says nothing about whether $\mathbf{No}^{M_1} \cong \mathbf{No}^{M_2}$ forces $M_1 = M_2$ for two inner models of a common universe, whose spectra could agree without settling their isomorphism problem (source 08).

12 A sharp theorem for bounded birthday fields (source 01)

The previous theorems concern all ground birthdays at once. A set-sized fragment has a new obstruction: its birthdays can become cofinally approachable by a short sequence, even if every individual sign of length below the bound remains old.

Let κ be an uncountable regular cardinal in M that remains a cardinal in N , and consider the set $(\mathbf{No}_{<\kappa})^M$. It is a real closed set field in M by the classical birthday-fragment theorem, and remains real closed in N by the finite-coefficient argument. Regularity in N is *not* assumed in advance.

Theorem 12.1 (Bounded-birthday preservation). *The following are equivalent.*

- (i) $(\mathbf{No}_{<\kappa})^M$ is κ -saturated in N as an ordered field.
- (ii) The order $(\mathbf{No}_{<\kappa})^M$ has the η_κ property in N .

- (iii) $\text{cf}^N(\kappa) = \kappa$ and $(^{\alpha}2)^N = (^{\alpha}2)^M$ for every $\alpha < \kappa$.
- (iv) κ is regular in N and $(\mathbf{No}_{<\kappa})^M = (\mathbf{No}_{<\kappa})^N$ as sets of sign sequences.

Proof. (i) $\Leftrightarrow$ (ii) is lemma 2.3(i), and (iii) $\Leftrightarrow$ (iv) is the representation of surreals by binary sign sequences.

Assume (iv), and let $L, R \subseteq (\mathbf{No}_{<\kappa})^M$ be N -sets with $L < R$ and $|L \cup R| < \kappa$. By regularity in N , $\gamma = \sup\{b(y) + 1 : y \in L \cup R\} < \kappa$. The cut constructed in N has birthday at most γ by (2.1), so it lies in $(\mathbf{No}_{<\kappa})^N = (\mathbf{No}_{<\kappa})^M$. This proves (ii).

For necessity, suppose first $\text{cf}^N(\kappa) = \mu < \kappa$. A cofinal sequence of ordinals below κ , viewed as all-plus surreals, is cofinal in the fragment by (2.2), has size below κ , and has no strict upper bound there; with an empty right side it violates (ii). Next suppose a new binary function of length below κ exists, and take one $s : \alpha \rightarrow \{-, +\}$ of least length. Its proper prefixes are old and have birthday below κ , so L_s, R_s of lemma 2.2 lie in the fragment and have size at most $|\alpha| < \kappa$. Any separator, even in $\mathbf{No}^N$, extends s ; an old separator cannot, since restricting it to α would put s in M . This violates (ii). $\square$

Remark 12.2 (Why the two criteria differ). In the full-class theorem, $\mathbf{NS}_{<\kappa}$ bans short sequences with *arbitrarily large range*. The fragment theorem bans new binary signs below a fixed birthday bound and separately preserves the cofinality of that bound. A short sequence with unbounded range at a much larger cardinal may be invisible to one small fragment but visible to the full old field. Conversely, singularizing the bound can destroy fragment saturation without adding any new sign below it (corollary 13.4). In the notation of part III, condition (iii) says $\text{cf}^N \kappa = \kappa$ and $\kappa \leq \beta(M, N)$ when $M \neq N$.

Proposition 12.3 (Pure complexification of a set fragment; source 01). *Let $\mu = |(\mathbf{No}_{<\kappa})^M|^N$. If μ is uncountable, the algebraically closed field $(\mathbf{No}_{<\kappa})^M[i]$ is μ -saturated in the pure field language in N , whether or not the conditions of theorem 12.1 hold.*

Proof. An algebraically closed field of characteristic zero and uncountable cardinality μ has transcendence degree μ over $\mathbb{Q}$: a basis of size τ would give an algebraic closure of size $\max(\tau, \aleph_0)$. Over fewer than μ parameters there is still a transcendental element, and algebraic types are realized by algebraic closedness; quantifier elimination gives μ -saturation. This is the usual dimension argument, included as a consequence rather than a new field theorem. $\square$

13 Forcing examples and distributivity

The theorems above apply to every pair of universes in their hypotheses. For set forcing they translate the usual preservation of short ordinal sequences into field-theoretic statements. The forcing facts used are standard [10, 11, 12, 13] and are not reproved.

13.1 The forcing-theoretic formulation

A forcing $\mathbb{P}$ is $< \kappa$ -distributive if the intersection of fewer than κ dense open subsets is dense, equivalently if its complete Boolean algebra is $< \kappa$ -distributive. With the usual separative interpretation this is equivalent to adding no new ordinal-valued sequences of length less than κ in any generic extension [10]; conventions about whether κ itself is included vary, and the sequence formulation is the one used here. For the forward direction, take for each coordinate of a name for a short sequence the dense set of conditions deciding it; a condition in the intersection decides the whole sequence as a ground sequence. Conversely, refine the dense open sets by maximal antichains in the complete Boolean algebra; a generic choice through them is a short

ordinal-coded sequence, and if all such sequences are old, conditions deciding a whole choice are dense below all the antichains. This is the standard Boolean form of the equivalence, not a new forcing theorem.

Corollary 13.1 (Forcing characterization; sources 01, 03, 04, 08). *Let κ be an uncountable cardinal of M and $\mathbb{P} \in M$.*

- (a) (source 03) *If κ is regular in M , then $\mathbb{P}$ adds no new ordinal sequences of length below κ in any generic extension if and only if, in every generic extension, $\mathbf{No}^M$ has the η_κ property. The preservation direction preserves κ and its regularity.*
- (b) (sources 04, 08) *If $\mathbb{P}$ preserves κ as an uncountable cardinal, the following are equivalent: $\mathbb{P}$ is $< \kappa$ -distributive in M ; in every generic extension $\mathbf{No}^M$ is externally κ -saturated; in every generic extension $(\mathbf{SC}^M, c_M)$ is externally κ -saturated.*

Without conjugation, $\mathbf{SC}^M$ is externally set-saturated in every generic extension, regardless of distributivity.

Proof. Apply (6.1), theorem 1.1, theorem 11.4 and theorem 11.2 in each generic extension. A new short cofinal map or a collapse to a smaller ordinal would be a forbidden short sequence, which gives the preservation clause in (a). $\square$

The phrase “in every generic extension” is indispensable. A property of one extension does not identify the behaviour of every condition of a nonhomogeneous forcing. For example, the lottery sum of trivial forcing and Cohen forcing has a generic branch that adds nothing and another that adds a real; the trivial branch preserves all saturation, but the whole forcing is not countably distributive (source 08). The invariants $\delta(M, N)$ and $\beta(M, N)$ belong to a particular pair of universes, not to the name of a poset, and the theorems are formulated pair by pair. One can also apply corollary 13.1 below a specified condition.

13.2 A Cohen real destroys even finite-parameter saturation

Let $N = M[g]$ add a Cohen real, for instance with $\mathbb{P} = \text{Fn}(\omega, 2, <\omega)$. The generic union is a new subset of ω : for every ground binary sequence the conditions disagreeing with it somewhere are dense. No finite ordinal sequence is new, so

$$\delta(M, N) = \beta(M, N) = \omega.$$

$\mathbf{No}^M$ has an (ω, ω) gap and is not even ω -saturated; the rational-cut type (7.1) witnesses this directly. The pure old order is ω -saturated but not ω_1 -saturated. Nevertheless $\mathbf{No}^M \preccurlyeq \mathbf{No}^N$ and $(\mathbf{SC}^M, c_M) \preccurlyeq (\mathbf{SC}^N, c_N)$ remain true, and the pure field $\mathbf{SC}^M$ remains saturated over every N -set; naming conjugation makes it fail ω -saturation again. Neither loss of real closedness nor failure of elementarity explains the omitted type: the change lies in which countable sets of formulas are available. Nothing here depends on Cohen combinatorics beyond adding a real.

13.3 Adding a subset of a regular cardinal

Proposition 13.2 (Any prescribed regular first gap; sources 01, 03, 04, 08). *Let κ be an infinite regular cardinal in M , and let $N = M[G]$ be obtained by $\text{Add}(\kappa, 1) = \text{Fn}_{<\kappa}(\kappa, 2)$, the partial functions $\kappa \rightarrow 2$ with domain of size below κ , ordered by reverse inclusion. Then*

$$\delta(M, N) = \beta(M, N) = \kappa.$$

$\mathbf{No}^M$ has an omitted (κ, κ) gap and no omitted pair of size below κ . If $\kappa > \omega$, no real is added, $\mathbf{No}^M$ and $(\mathbf{SC}^M, c_M)$ are ω -saturated and κ -saturated, and they are not $(\kappa^+)^N$ -saturated. No cardinal arithmetic assumption is needed.

Proof. For uncountable κ a descending chain of fewer than κ conditions has its union as a lower bound, by regularity. To decide a name for an ordinal-valued sequence of length $\lambda < \kappa$, strengthen a condition coordinate by coordinate, taking unions at limits and at the end; the result decides the sequence as a ground sequence. So no such sequence is added, and κ stays a regular cardinal. The generic union $g : \kappa \rightarrow 2$ is total and new: given an old h , it is dense to put the opposite value at an unused coordinate. Hence $\delta = \kappa$. No subset of an ordinal below κ is added, and g codes a new subset of κ , so $\beta = \kappa$. Apply theorems 6.2, 7.1, 9.2 and 11.4 and corollary 6.4. For $\kappa = \omega$ this is the Cohen case. The generic sign of length κ is itself fresh, so its gap can also be read from theorem 8.4. $\square$

No assumption such as $2^{<\kappa} = \kappa$ is needed for this first-gap calculation; the saturation failure is indexed by the successor computed in the actual extension. The proposition does not compute the full gap spectrum, does not claim that no larger gaps occur, and asserts nothing about the preservation of cardinals above κ . For $\kappa = \omega_1^M$, no reals are added, ω - and ω_1 -saturation survive, and ω_2^N -saturation is lost: failures at different parameter sizes.

Example 13.3 (Under $\kappa^{<\kappa} = \kappa$; source 04). If moreover $M \models \kappa^{<\kappa} = \kappa$ with κ uncountable, then $|\text{Add}(\kappa, 1)| = \kappa$, so the forcing has the κ^+ -chain condition. Together with closure this preserves κ , κ^+ and the regularity of κ [10], so the failure of saturation occurs at the ground model's κ^+ . This hypothesis is stronger than needed for proposition 13.2; it is recorded because source 04 uses it for this extra preservation clause.

13.4 Prikry forcing separates the two countable notions

Assume M has a measurable cardinal ρ , and force with classical Prikry forcing from a normal measure on ρ . Its standard properties are that it preserves cardinals, adds an increasing cofinal sequence $g : \omega \rightarrow \rho$, changes $\text{cf}(\rho)$ to ω , and adds no bounded subsets of ρ ; in particular it adds no reals [11, 12, 13]. These are imported facts, not reproved here. The sequence g is new: M regards ρ as regular and uncountable. The range of g is a new subset of ρ , since in M its order type ω and cofinality in ρ would contradict regularity. Consequently

$$\delta(M, N) = \omega, \quad \beta(M, N) = \rho, \quad \mathbb{R}^M = \mathbb{R}^N. \quad (13.1)$$

Corollary 13.4 (Prikry separation; sources 01, 03, 04, 08). *In this Prikry extension:*

- (i) $\mathbf{No}^M$ and $(\mathbf{SC}^M, c_M)$ are ω -saturated but not ω_1 -saturated, while $\mathbf{SC}^M$ is set-saturated.
- (ii) $(\mathbf{No}_{<\rho})^N = (\mathbf{No}_{<\rho})^M$, yet this set field is not ρ -saturated in N : its all-plus ordinals acquire a countable cofinal sequence (source 01).
- (iii) Applying the block code to g gives an omitted (ω, ω) gap whose simplest new separator code(g) has birthday exactly ρ (source 08).
- (iv) The characteristic sign s of $\text{ran}(g)$, of length ρ , is fresh, and its gap has cofinalities $(\text{cf}^N \rho, \text{cf}^N \rho) = (\omega, \omega)$, not (ρ, ρ) (source 03).
- (v) The endpoints of the omitted cut obtained from the absolute interval code are the finite rational expressions of proposition 5.5 with $\alpha_j = g(j)$ (source 04).

Proof. (i) Theorem 7.1 gives ω -saturation, the new countable sequence and theorem 1.1 give failure at ω_1 , and theorems 11.2 and 11.4 give the surcomplex clauses. (ii) Every binary sign of length below ρ codes a bounded subset of ρ ; the regularity clause of theorem 12.1 fails. (iii) Every finite initial segment of g is old, so theorem 5.8 applies with $\lambda = \omega = \delta$; $b(\text{code}(g)) = \sum_{n < \omega} (g(n) + 2) = \rho$, since the finite partial sums lie below the cardinal ρ and are cofinal in it. (iv) Every proper initial segment of $\text{ran}(g)$ is finite, so every proper prefix of s is old, and $\text{ran}(g)$ is new; apply lemma 8.3. (v) is proposition 5.5. $\square$

The parameters of the omitted cut all have birthdays below ρ and were all present in M ; what is new is their countable arrangement. The alphabet in the interval tree must be allowed to reach ρ : a test restricted to new reals, or to new binary signs of countable length, would miss this example. The old universe's cofinality of the birthday ρ is the wrong cofinality for the gap computation. Corollary 13.4(ii) shows directly why regularity is an independent clause in theorem 12.1, and (iii) shows how far $\beta = \rho$ can lie above $\delta = \omega$.

13.5 Comparison table

Extension	δ	β	New reals?	Old ordered field $\mathbf{No}^M$	Old pure $\mathbf{SC}^M$
Cohen real	ω	ω	Yes	Not ω -saturated	Set-saturated
Add($\kappa, 1$), $\kappa > \omega$ regular	κ	κ	No	κ -saturated, not $(\kappa^+)^N$ -saturated	Set-saturated
Prikry at a measurable ρ	ω	ρ	No	ω -saturated, not ω_1 -saturated	Set-saturated

Adding conjugation to the last column reproduces the ordered-field column. The table describes external saturation in the stated outer universe; it does not alter the internal theorem that each universe's own surreal field realizes all of that universe's set-sized cuts.

14 Why order gaps do not become Cauchy limits

The native fine topology of a surreal field uses all positive radii of that field, not only real ones; its additive uniformity has the entourages $|x - y| < \varepsilon$ for $\varepsilon \in \mathbf{No}_{>0}^M$. On the surcomplex field use either the surreal-valued modulus $|a + bi| = \sqrt{a^2 + b^2}$ (source 01) or $\|a + bi\|_\infty = \max(|a|, |b|)$ (source 04); they define the same product topology and uniformity. All claims are about neighborhoods and set-indexed families; no set of all open classes is formed.

Proposition 14.1 (External discreteness and Cauchy rigidity; sources 01, 03, 04). *For every pair $M \subseteq N$ under (H) :*

- (i) *every N -set $A \subseteq \mathbf{No}^M$ is closed in the fine topology of $\mathbf{No}^M$ and uniformly discrete: some old $\varepsilon > 0$ is below every nonzero distance in A ;*
- (ii) *every N -set-indexed Cauchy net in $\mathbf{No}^M$ is eventually constant, and every convergent such net is eventually equal to its limit;*
- (iii) *the same holds for $\mathbf{SC}^M$ with either surcomplex modulus.*

No short-sequence hypothesis is needed.

Proof. The nonzero pairwise distances in A form an N -set of positive old numbers. Theorem 3.5 with left side $\{0\}$ (source 01), or directly lemma 3.3 applied to the reciprocals (source 04), gives a positive old lower

bound ε . For $x \in \mathbf{No}^M \setminus A$ apply the same argument to $\{|x - a| : a \in A\}$ to get a neighborhood of x missing A . The range of a set-indexed net is an N -set; choosing ε below all nonzero pairwise distances in the range, the Cauchy condition for that one entourage makes late values equal. For a convergent net, apply the separation argument to its range together with its limit. For $\mathbf{SC}^M$, the modulus of an old nonzero number is old and positive; apply the same argument. $\square$

For the interval tree, the widths $b_{f \upharpoonright \alpha} - a_{f \upharpoonright \alpha}$, $\alpha < \lambda$, are positive old numbers with a common positive old lower bound; the endpoint sequences are not Cauchy in the fine uniformity, and the omitted cut is not evidence of a missing fine-Cauchy limit. Even a Prikry cut with countably many endpoints leaves the proposition intact. A topology that looks only at selected scales could behave differently, but it is a different structure and is not used. Elementarity, saturation and fine completeness must not be treated as interchangeable: the old field remains real closed and elementary, loses specified types, and keeps this set-indexed rigidity.

Lemma 14.2 (Prefix cones are open; source 04). *For every $s \in \mathbf{No}^N$, $\text{Cone}(s)$ is open in the fine order topology of $\mathbf{No}^N$.*

Proof. By lemma 2.2, $\text{Cone}(s)$ is the class of separators of $L_s < R_s$. For $x \in \text{Cone}(s)$, the set $\{x - l : l \in L_s\} \cup \{r - x : r \in R_s\}$ of positive distances has a positive lower bound in $\mathbf{No}^N$ by the cut theorem, and a smaller interval about x keeps all the canonical inequalities. If the set is empty, $\text{Cone}(s) = \mathbf{No}^N$. $\square$

Theorem 14.3 (Closed, nowhere dense, and still set-discrete; sources 03, 04). *$\mathbf{No}^M$ is closed in $\mathbf{No}^N$, and its subspace topology is its own order topology. If $M \neq N$, $\mathbf{No}^M$ is nowhere dense in $\mathbf{No}^N$. The same statements hold for $\mathbf{SC}^M \subseteq \mathbf{SC}^N$ in the product topologies.*

Proof. If $x \in \mathbf{No}^N \setminus \mathbf{No}^M$, every extension of x is new, because an old sign has only old prefixes; so the open cone $\text{Cone}(x)$ misses $\mathbf{No}^M$. Source 03's route uses the open interval $(x \frown -, x \frown +)$, all of whose points extend x .

For every positive $r \in \mathbf{No}^N$ there is an old $0 < e < r$ (lemma 3.3); this gives equality of the intrinsic and subspace topologies at every old point. Assume $M \neq N$ and choose a new z ; replacing it by $h = |z|/(1 + |z|)$ gives a new $h \in (0, 1)$, since otherwise $|z| = h/(1 - h)$ would be old. Given $a \in \mathbf{No}^M$ and $r > 0$, the point $a + eh$ lies in $(a - r, a + r)$ and is new, since otherwise $h = ((a + eh) - a)/e$ would be old. So no old point is interior; a closed class with empty interior is nowhere dense. (Source 03 instead uses $a + z/\gamma$ for an old ordinal $\gamma > |z|/r$.) For the complex fields use $\mathbf{No}^M \times \mathbf{No}^M \subseteq \mathbf{No}^N \times \mathbf{No}^N$: products of closed coordinate classes are closed, the radius comparison gives the subspace claim, and perturbing one real coordinate by eh shows empty interior. $\square$

The old field is therefore simultaneously cofinal, simplicity-initial, elementary, closed and nowhere dense in $\mathbf{No}^N$. Order-cofinality must not be mistaken for topological density, nor elementary inclusion for density. In the outer field the whole interval of fillers of a fresh gap, the cone of theorem 8.5, is separated from every old point. No convergence of an ordinary sequence to a missing point is asserted.

Remark 14.4 (Relation to the foundations report). The internal statement is Theorem 12.1 (`found:thm:discrete`) of `docs/foundations-and-computation/foundations`, which proves (i)–(iii) of proposition 14.1 for sets of the ambient universe, assuming its Lemma 3.3 (`found:lem:bounds`, `bounds` and `small positive lower bounds`) “for the permitted small families”. Proposition 14.1 extends it to *new* N -sets of old numbers. Its only input is the existence of old positive lower bounds, which holds in every same-ordinal extension even where cut filling fails. In the Lean ledger (`docs/FORMALIZATION.md`), the Cauchy-net clause of `found:thm:discrete` is stated under the hypothesis `HasSmallCutFillers`, that is, under `small cut`

filling. The external version shows that positive lower bounds suffice for that clause. This is recorded as information for formalization; nothing here is formalized.

The foundations report’s Section 7.7 (`found:sub:countable`) warns: “‘every set family has a bound’ means every family represented by a set *in the relevant model*; it is not an absoluteness claim about every family visible from a stronger metatheory.” That warning concerns countable models and different ordinals, and it stands. For same-ordinal outer universes this report sharpens it: bounds of outer sets of old numbers *are* absolute (lemmas 3.2 and 3.3), while cut filling is not (theorem 1.1). The different-height analogue, the embedding $\mathbf{No}_U \rightarrow \mathbf{No}_V$ for Grothendieck universes, is treated in that report and is not used here. The catalogue’s remark that universe-relative formulations “require smaller-universe subsets and index types” also concerns carriers of different heights; in the present setting the index sets need not be ground-small.

Part V

Class isomorphisms and real forms (Global Choice)

15 Pure-field class isomorphisms

Throughout this part assume (GC): the ambient theory is GBC with Global Choice, and M is a class of the class universe. Source 04 works with a definable M and definable fields, which is a special case. All assertions concern individual class maps; $\text{Aut}(\cdot)$ is shorthand for quantification over class field automorphisms, and no class-object of all class maps is formed. At every ordinal stage of the extension arguments, the partial fields and maps are sets; their histories below a given ordinal are sets by class Replacement, and deterministic choices from a fixed set-like global well-order permit set-valued transfinite recursion with class parameters and its class union. No general recursion with class-valued stages, no class Zorn lemma and no unrestricted class truth predicate is assumed.

Theorem 15.1 (Class back-and-forth; prior result). *Let H_1, H_2 be proper-class algebraically closed fields of the same characteristic. Every isomorphism between set-sized subfields of H_1 and H_2 extends to a class field isomorphism $H_1 \rightarrow H_2$.*

This is Theorem 9.1 (`saut:thm:acfhomogeneity`) of `docs/surcomplex/surcomplex-field-automorphisms`, with the same hypothesis (Global Choice). Sources 03 and 04 reprove it by the same route: enumerate both fields along Ord by set-like global well-orders, extend at a forth step by a root of the transported irreducible polynomial or by an element transcendental over the current target field — which exists by lemma 11.1 and proper-class size — do the symmetric back step, and take unions at limits. Source 03 included the proof because applying a theorem stated for set-sized fields to proper-class fields without such a check would leave a foundational gap; the collection’s statement is already for proper-class fields.

Second route (source 08). Let H_1, H_2 be proper-class algebraically closed fields of characteristic zero that are saturated over every set of parameters. Every set-sized partial elementary map $H_1 \rightarrow H_2$ extends to a class isomorphism. At a forth step, transport the complete type of the next uncovered element over the current domain, a set by quantifier elimination; it is finitely satisfiable in H_2 by partial elementarity, and set-saturation of H_2 realizes it, outside the current range because the type records inequality with every element of the domain. Back steps are symmetric, and the least realization in a fixed global well-order makes each step specified. The method is classical [6, 7].

Theorem 15.2 (Old and new pure surcomplex fields; sources 03, 04, 08). *For every N -set $A \subseteq \text{SC}^M$ there is a class field isomorphism $\Phi : \text{SC}^M \rightarrow \text{SC}^N$ fixing A pointwise. If $M \neq N$, no such isomorphism maps $\mathbf{No}^M$ onto $\mathbf{No}^N$, and none satisfies $\Phi \circ c_M = c_N \circ \Phi$.*

Proof. Both fields are proper classes, since each contains every surreal ordinal in its real coordinate, and are algebraically closed in N (proposition 3.1). The identity on the field generated by A is an isomorphism of set-sized subfields; apply theorem 15.1. (Along the second route: the identity on A is partial elementary by proposition 3.1, both fields are set-saturated by theorem 11.2.) If $\Phi[\mathbf{No}^M] = \mathbf{No}^N$, its restriction would be an isomorphism of real closed fields, contrary to corollary 11.6; class Replacement supplies the set-image condition. An isomorphism intertwining the conjugations would carry one fixed field onto the other. $\square$

The inclusion $\text{SC}^M \hookrightarrow \text{SC}^N$ and the isomorphism Φ are different maps: for $M \subsetneq N$ the inclusion is a proper embedding, and the existence of Φ does not make it surjective. Φ is noncanonical. It need not preserve old coordinates, the natural valuation, the real axis, simplicity, an exponential, or a derivation, and no such compatibility is claimed. Theorem 15.2 settles, under Global Choice, source 01's question on class isomorphisms $\text{SC}^M \cong \text{SC}^N$ (section 19).

16 Gap invariants of surcomplex real forms

A *real form* of an algebraically closed field H of characteristic zero is a real closed subfield $F \subseteq H$ with $H = F(i)$; its conjugation fixes F and sends i to $-i$. This is the usage of the automorphism report (`saut:sec:realforms`). For a real closed class field F , let $\text{GapSpec}_N(F)$ be the class of cofinality pairs of its set-presented gaps, and let $g_N(F)$ be the least size of an omitted separated N -set pair, with the bookkeeping value $g_N(F) = \infty$ (not a cardinal) when every set pair is filled.

Lemma 16.1 (Invariance; source 03). *A field isomorphism between real closed class fields preserves their gap spectra and least omitted-pair sizes. Two involutions of SC^N are conjugate by a class field automorphism if and only if their fixed real fields are isomorphic.*

Proof. A field isomorphism preserves nonzero squares and hence the unique order; its restriction to an N -set is a set bijection by class Replacement, so it preserves cardinalities, cofinalities and the existence of separators. The second statement is the conjugacy criterion of `saut:sec:realforms`: restrict a conjugating automorphism, or extend $f : F_1 \rightarrow F_2$ by $x + iy \mapsto f(x) + if(y)$. It is the transport mechanism, not a new invariant. $\square$

Theorem 16.2 (A forcing-sensitive real form without set-sized cofinal subsets; sources 03, 04, 08). *Let $M \subsetneq N$, choose $\Phi : \text{SC}^M \rightarrow \text{SC}^N$ by theorem 15.2, and put*

$$j = \Phi c_M \Phi^{-1}, \quad H = \Phi(\mathbf{No}^M).$$

Then j is a field involution of SC^N with fixed field H and $\text{SC}^N = H(i)$, and

$$\text{GapSpec}_N(H) = \{(\text{cf}^N \mathbf{b}(s), \text{cf}^N \mathbf{b}(s)) : s \in \text{Fresh}(M, N)\}, \quad (16.1)$$

$$g_N(H) = \delta(M, N). \quad (16.2)$$

H has no N -set cofinal or coinital subset, and every N -set of elements of H is bounded above and below in H . Nevertheless j is not conjugate to c_N by any class field automorphism of SC^N .

Proof. Transport of structure gives $\text{Fix}(j) = \Phi(\text{Fix}(c_M)) = H$ and $\text{SC}^N = H(i)$; even if $\Phi(i) = -i$ the quadratic extension is unchanged. The displayed equations follow from corollary 8.6, theorem 9.2, and lemma 16.1. A cofinal N -set in H would pull back to a cofinal set in $\mathbf{No}^M$, contrary to lemma 3.3; the same holds for coinitial sets and for bounds. Finally $\mathbf{No}^N = \text{Fix}(c_N)$ fills every N -set cut, so $g_N(\mathbf{No}^N) = \infty \neq \delta(M, N) = g_N(H)$; the fixed fields are not isomorphic, and lemma 16.1 gives non-conjugacy. $\square$

Theorem 16.3 (Elementarily equivalent, not conjugate; source 04). *In the situation of theorem 16.2:*

- (i) $\text{Fix}(c_N) = \mathbf{No}^N$ and $\text{Fix}(j)$ is isomorphic as an ordered field to $\mathbf{No}^M$; both fixed fields are real closed, and $\text{SC}^N = \text{Fix}(c_N)(i) = \text{Fix}(j)(i)$;
- (ii) c_N and j are not conjugate by any class field automorphism;
- (iii) (SC^N, c_N) and (SC^N, j) are elementarily equivalent, but their external saturation spectra differ: for uncountable κ the second is κ -saturated exactly when $\kappa \leq \delta(M, N)$, whereas the first is set-saturated; at ω the second is saturated exactly when $\mathbb{R}^M = \mathbb{R}^N$.

Proof. (i) and (ii) are theorem 16.2; source 04 derives (ii) from corollary 11.6. Both expansions are interpreted, with the same coordinate rules, in real closed fields, and RCF is complete, so they are elementarily equivalent. Their spectra are those of their fixed fields by theorem 11.4 transported along Φ , and corollary 6.4, theorem 7.1, and lemma 2.3(iii) compute them. $\square$

Three statements must not be confused: the embedded real axes are different; the real axes are not isomorphic as fields; and the involutions are not conjugate. Applying a phase automorphism to one real axis proves the first but produces a *conjugate* involution and an isomorphic fixed field. The nonisomorphism is detected by external types, not by a first-order sentence.

Remark 16.4 (What is new relative to the automorphism report). Proposition B.1 (saut:prop:conjugacycriterion) of docs/surcomplex/surcomplex-field-automorphisms states that an involution of $\mathbf{No}[i]$ is conjugate to c if and only if its fixed real field has the set-cut interpolation property. It was present at the sources' pin as prop:conjugacycriterion, and none of the four sources cites it. Read in N , its easy direction (the property holds in $\mathbf{No}^N$ by Conway cuts and is preserved by ordered isomorphisms) already shows that j is not conjugate to c_N : $\text{Fix}(j) \cong \mathbf{No}^M$ omits an N -set cut whenever $M \neq N$ (theorem 1.1). The non-conjugacy clauses of theorems 16.2 and 16.3 are therefore consequences of that proposition together with the spectrum theorem. What is genuinely new in this part is:

- (a) *no set-sized cofinal subset*: the transported real forms, like $\mathbf{No}^N$ itself, have no set-sized cofinal subset, so the cofinality obstruction of the automorphism report's Theorem 10.2 (saut:thm:inequivalentrealform, a real form with a countable cofinal subset) cannot separate them; the obstruction lies inside a bounded cut;
- (b) *several forms separated from one another* by the invariant g_N and the gap spectrum (theorem 16.6), which the interpolation criterion alone does not do, since it only separates a form from c ;
- (c) *agreement with c_N on ordinary complex numbers* (corollary 16.5);
- (d) *elementary equivalence with different external spectra* (theorem 16.3), and the exact values of the spectra.

These continue, but do not settle, the automorphism report's open "real-form and continuity classifications" (its Section 14): no classification of real forms or involutions is obtained.

Corollary 16.5 (Agreement on prescribed ordinary data; source 03). *Let $A \subseteq \text{SC}^M$ be a set-sized subfield stable under c_M (as a subfield of SC^N it is then stable under c_N). The involution j can be chosen to agree with c_N on every element of A . If $\mathbb{R}^M = \mathbb{R}^N$, it can be chosen to agree with c_N on all of $\mathbb{C}^N$.*

Proof. Choose Φ fixing A pointwise. For $a \in A$, $c_M(a) \in A$, so $j(a) = \Phi(c_M(\Phi^{-1}(a))) = \Phi(c_M(a)) = c_M(a) = c_N(a)$. When no reals are added, $\mathbb{C}^M = \mathbb{C}^N$ is a set-sized common subfield stable under conjugation. $\square$

Thus agreeing with ordinary conjugation on every ordinary complex constant does not identify the surreal real axis even up to conjugacy. The obstruction is not an unusual choice of sign for i or a moved ordinary coefficient.

16.1 Several ground real forms in one field

Theorem 16.6 (Different thresholds separate real forms; sources 03, 04). (i) (source 03) *Let $r \geq 1$, and suppose that the models M_a , $a < r$, are proper transitive inner models of N with the same ordinals, each available as a class and satisfying the standing hypotheses, and that the cardinals $\delta(M_a, N)$ are pairwise distinct. Then SC^N has $r + 1$ pairwise nonconjugate field involutions, including c_N , none of whose fixed fields has a set-sized cofinal or cointial subset. If each M_a has the same reals as N , all of them can be chosen to agree with c_N on $\mathbb{C}^N$.*

(ii) (source 04) *Suppose $M_1, \dots, M_n$ satisfy the standing hypotheses and their old surreal fields have pairwise different external saturation spectra in N . Then SC^N has n pairwise nonconjugate real-closed-field involutions realizing those spectra, and c_N supplies one more whenever none of the listed spectra is the full set-saturation spectrum.*

Proof. Apply theorem 16.2 to each M_a with its own isomorphism $\Phi_a : \text{No}^{M_a}[i] \rightarrow \text{SC}^N$. The fixed fields have least omitted-pair sizes $\delta(M_a, N)$, and that of c_N is ∞ ; all values differ, so lemma 16.1 separates the conjugacy classes. Cofinality and agreement come from theorem 16.2 and corollary 16.5. For (ii), a conjugacy between two transported involutions would give an isomorphism of fixed fields, which preserves external saturation; No^N is set-saturated. $\square$

Part (ii) realizes a *given* finite family of inner models; it does not assert that arbitrary families of saturation spectra can be produced by forcing without further hypotheses.

Corollary 16.7 (Arbitrarily long finite lists; source 03). *For every positive integer r there are finite forcing-tower constructions whose final surcomplex field admits $r + 1$ involutions as in theorem 16.6(i), all agreeing on every ordinary complex number. No large-cardinal assumption is needed. This is a relative forcing construction over a suitable ambient model, not an assertion that every universe contains the requisite grounds.*

Proof. Starting from M_0 , build $M_0 \subset M_1 \subset \dots \subset M_r = N$: at stage a choose in M_a an uncountable regular cardinal κ_a above all previously chosen ones, and force with $\text{Add}(\kappa_a, 1)$. Each later stage is closed under decreasing sequences of length below its own larger regular cardinal, so it adds no ordinal sequences of length below κ_a and preserves κ_a as a regular cardinal: a sequence in the final model of length below κ_a was present before the last forcing by closure, and repeating through the finitely many later stages puts it in M_a . The generic subset of κ_a added at stage a remains new over M_a in N . Thus $\delta(M_a, N) = \kappa_a$, pairwise distinct in N . No stage adds reals, so all the models have the same ordinary complex field. Apply theorem 16.6(i). $\square$

The finite tower is intentional: an infinite iteration needs support and preservation arguments not supplied by the finite proof, and no claim is made that an arbitrary proper class of gap invariants can be realized simultaneously.

Part VI

A Boolean completion of reverse simplicity (source 01)

17 A Boolean completion of reverse simplicity

Berenbeim’s Question 2 asks for a complete class Boolean algebra in which the reverse-simplicity order of surreal numbers embeds densely [15, p. 32]. The answer depends on the meaning of “complete.” We give an explicit positive answer for joins of sets, and a negative answer for joins of all subclasses under Global Choice. This section works in a single universe and does not use the pair $M \subseteq N$.

This is a different use of the sign tree from numerical cuts: *stronger* means extending a sign, regardless of its numerical value. Identifying $-$ and $+$ with 0 and 1, the partial order is

$$P = {}^{<\text{Ord}}2, \quad p \leq_P q \iff q \subseteq p.$$

Each condition is a set, and P is a proper class; it is the reverse of the non-strict initial-segment simplicity order.

17.1 A direct limit with set-coded elements

For each ordinal α let $B_\alpha = \mathcal{P}({}^\alpha 2)$, and for $\alpha \leq \gamma$ define the cylinder embedding

$$j_{\alpha\gamma}(A) = \{s \in {}^\gamma 2 : s \upharpoonright \alpha \in A\}. \quad (17.1)$$

These are injective Boolean homomorphisms preserving arbitrary set-indexed unions and intersections, with $j_{\gamma\epsilon}j_{\alpha\gamma} = j_{\alpha\epsilon}$. Identify pairs (α, A) with $A \subseteq {}^\alpha 2$ when their images agree at a common later level. A proper-class equivalence class must not be made an element; instead represent each class by its unique *least-level pair*. Among the levels at most a representative’s level there is a least one from which it is a cylinder, and injectivity makes its subset unique. Taking the least-level pair is a definable operation on set-coded representatives.

Let $\mathbb{B}$ be the class of canonical pairs. To perform a finite Boolean operation, lift the arguments to a common level, operate in the power set there, and take the canonical pair; the result is independent of the common level by (17.1). Thus $\mathbb{B}$ is a class Boolean algebra whose elements are sets.

Theorem 17.1 (Set-complete dense completion). *$\mathbb{B}$ is complete for every set-sized family of its elements. The map*

$$e : P \longrightarrow \mathbb{B} \setminus \{0\}, \quad e(p) = [(b(p), \{p\})],$$

is an injective dense order embedding; in particular $e(p) \leq_{\mathbb{B}} e(q) \iff q \subseteq p$.

Proof. Let S be a set of canonical pairs and choose an ordinal γ at least every level occurring in S . Lift all of them to B_γ and take the union; its canonical pair is their join, since any upper bound can be moved to a common later level where the power-set union lies below it, and cylinder embeddings preserve that union. Complements give meets; the empty join and meet are 0 and 1.

A condition p determines its nonempty cylinder of extensions. If p extends q , its cylinder lies in that of q ; if they disagree, the cylinders are disjoint; if p is a proper prefix of q , the cylinder of p is not contained in that of q (extend p with the opposite sign at the next coordinate of q). This gives the order equivalence and injectivity. Finally, if $b \neq 0$ is represented by (α, A) with $A \neq \emptyset$ and $p \in A$, then $e(p) \leq b$, so the embedding is dense. $\square$

The construction answers the existence question when “complete” means set-complete, the size convention in which a class Boolean algebra has joins of its set-sized subsets, as in Holy–Krapf–Lücke–Njegomir–Schlicht [16]. General existence and uniqueness questions for class forcing are subtle; the construction works directly for this binary tree. It is not claimed that every class partial order admits the same construction, nor that this completion is the unique class completion.

17.2 Why joins of all subclasses are impossible

For the negative statement only, work in GBC with Global Choice. A class Boolean algebra is *class-complete* if every subclass of its carrier, including proper subclasses, has a supremum in the carrier. This is strictly stronger than set-completeness.

Lemma 17.2 (Diagonal obstruction). *In GBC with Global Choice, a class Boolean algebra with an ordinal-indexed proper-class antichain of nonzero elements is not class-complete.*

Proof. Let $(a_\alpha)_{\alpha \in \text{Ord}}$ be such an antichain. The carrier is a proper class, and Global Choice gives a class surjection $h : \text{Ord} \rightarrow \mathbb{B}$ (restrict a set-like global well-order). Let $A = \{\alpha : a_\alpha \not\leq h(\alpha)\}$. If $\{a_\alpha : \alpha \in A\}$ had a supremum b , then

$$a_\alpha \leq b \iff \alpha \in A. \quad (17.2)$$

Indeed, for $\alpha \notin A$, a_α is disjoint from every member of the subclass, so $\neg a_\alpha$ is an upper bound and $b \leq \neg a_\alpha$; since $a_\alpha \neq 0$, $a_\alpha \not\leq b$. The other direction is the definition of supremum. Choose γ with $h(\gamma) = b$; then $\gamma \in A \iff a_\gamma \not\leq b \iff \gamma \notin A$, a contradiction. All classes used are legitimate comprehension instances; no collection of all classes is formed. $\square$

Theorem 17.3 (No all-subclass completion). *In GBC with Global Choice there is no class-complete Boolean algebra with a dense order embedding of $P = {}^{<\text{Ord}}2$ into its nonzero part.*

Proof. The signs $p_\alpha = +^\alpha \frown -$, $\alpha \in \text{Ord}$, are pairwise incompatible. Under a dense embedding their images are pairwise disjoint, since a nonzero meet would contain the image of a condition, a common extension of incompatible signs. They form an ordinal-indexed proper-class antichain; apply lemma 17.2. $\square$

The two answers are consistent: a set-sized family of cylinders has a common level at which its join is computed, and a proper-class family need not. Set-completeness neither implies nor purports to imply all-subclass completeness. This also prevents the misleading claim that Question 2 requires an inconsistency of ordinary set or class theory.

Part VII

Scope, questions and limits

18 Scope and significance

18.1 What the new information consists of

The usual statement that the full surreal class realizes every set-sized cut is internal to a universe. The main theorem explains exactly how it changes when the elements are held fixed while the allowed families of parameters change. Its most distinctive features are these:

1. a cut with one actual ground side is filled even if its other side is a new and arbitrarily large set;
2. a first new sequence can be encoded by a cut of the same cardinal length, however large its ordinal range;
3. every set-presented failure of the cut principle comes from a unique fresh sign, and is symmetric;
4. removing the distinguished real structure by passing to the pure surcomplex language erases the obstruction entirely.

Together they identify a concrete set-theoretic invariant inside the external order of the old surreal field, and show that “the model theory of surcomplex numbers” has no language-independent saturation answer.

The interval method (source 01) uses only set-sized interpolation inside the ground order, enough room to split an interval into any prescribed set of pairwise disjoint subintervals, and a ground-set containment property for external parameter sets. Surreal numbers supply all three explicitly. Other definable universal orders with the same properties may be worth studying; no classification of such orders is claimed.

18.2 Limits of the assertions

The argument does not address exponential or differential expansions of **No**, the Berarducci–Mantova derivation, or expansions naming simplicity. Their types are not classified by real closed cuts, so the sufficiency proof cannot simply be copied after adding those operations; even absoluteness of the individual old operations would not give a type-realization theorem in the expanded language. An elementary reduct can be well behaved while an expansion has additional omitted types [8, 7].

The results do not assert that every class function or arbitrary proper-class cut is available in ZFC. Set-sized data, the ground predicate, and the stated uses of class principles are hypotheses, not conventions to be supplied later. The report is not a consistency proof for ZFC, GBC, measurable cardinals or the model configurations; the Prikry examples are conditional on a measurable cardinal. No countable set model whose ordinals stop at an external height is used; the same-ordinals hypothesis is essential to the birthday cover.

No theorem here is Lean-checked. The collection’s existing formal proofs and the source manuscripts are separate evidence. Successful typesetting is not mathematical verification, and the finite Python checks (section C) verify no transfinite claim.

18.3 A formalization route without class-sized type objects

These are proposed proof tasks, not existing declarations. A first target is the branch-decoder theorem: a set ordinal, a set alphabet, a table of intervals, and an abstract predicate for old sequences; the obligations are

sibling uniqueness, limit nesting, totality of the default decoder and its agreement with an externally coded branch (source 01). Source 03 proposes starting instead with a generic ordered sign tree and an initial subclass predicate, proving lemmas 8.2 and 8.3 and theorem 8.4 as ordinal-combinatorial statements. The one-old-side theorem needs a bounded-stage cover and an internal cut operation, and should quantify over actual set-sized sides. For a universe-polymorphic proof assistant, the fragment theorem (theorem 12.1) is the cleanest end-to-end target: all structures can live in a large universe while the two set models remain distinct; full-class formulations become metatheorems or universe-relative schemes, not an illicit same-universe type of all surreals. The class back-and-forth should be formalized with set-stage maps and a class-theory interface. The surcomplex results separate into two reusable lemmas: external type realization for a proper-class algebraically closed field with ground-set parameter covers, and interpretation of the ordered base field through fixed squares in the quadratic complexification, which avoids repeating the collection’s quadratic-algebra development. The relation of proposition 14.1 to the Lean form of `found : thm : discrete` is recorded in remark 14.4.

19 Questions, and which of them the merge answered

The sources’ questions are recorded here with their status. They are directions suggested by the results, not claims that a particular published or catalogued question remains open.

Question 19.1 (External spectra of expansions; sources 01, 08; open). Determine the external saturation spectrum of old surreal exponential fields, and more generally of canonical expansions (surreal exponentiation, a specified derivation). What extra obstruction appears after naming them?

The main theorem gives a necessary condition, since the ordered-field reduct must be saturated; no sufficiency theorem for an expanded language is proved.

Question 19.2 (The cut-cofinality spectrum above δ ; source 01; asymmetric part answered by source 03). Source 01 asked to classify more of the external cut-cofinality spectrum than its first threshold, and to decide which asymmetrical pairs can occur.

Status. The asymmetric part is answered by source 03: every set-presented gap is symmetric, and the spectrum is exactly $\{(\text{cf}^N \mathbf{b}(s), \text{cf}^N \mathbf{b}(s)) : s \in \text{Fresh}(M, N)\}$ (corollary 8.6); source 01’s “set-character gaps” are source 03’s “set-presented gaps”. What stays open is which symmetric pairs above δ occur for a specified forcing, that is, computing the outer cofinalities of the lengths of fresh signs; proposition 13.2 computes only the first gap. This is related to, and must not be conflated with, the fresh-function spectra of Fischer–Koelbing–Wohofsky [14], whose spectrum-realization questions are not resolved here.

Question 19.3 (Class isomorphisms of the pure surcomplex fields; source 01; settled under Global Choice). Under which weak class-theoretic assumptions is there a class isomorphism $\text{SC}^M \cong \text{SC}^N$?

Status. Under GBC with Global Choice there is one, even fixing any N -set of old parameters: theorem 15.2 (sources 03, 04, 08), by the collection’s class back-and-forth (`saut : thm : acfhomogeneity`). Only the question whether an assumption *weaker* than Global Choice suffices remains open. Set-saturation (theorem 11.2) gives local realizations but is not itself a global construction.

Question 19.4 (Completeness of the invariant; sources 03, 08; open, one part answered). For same-ordinal inner models M_0, M_1 of a fixed N , which invariants beyond the least omitted-cut size distinguish $\mathbf{No}^{M_0}$ and $\mathbf{No}^{M_1}$ as class orders or real closed fields in N ? Under what hypotheses does equality of $\text{GapSpec}_N(\mathbf{No}^{M_0})$ and $\text{GapSpec}_N(\mathbf{No}^{M_1})$ imply isomorphism?

Status. Source 08’s remark that its construction “provides one family of gap witnesses, but not a complete spectrum of all omitted cut characters” is answered for a fixed pair by source 03’s corollary 8.6. The

classification question stays open: the spectrum records which cofinalities occur, not how many gaps of each kind occur or how they are arranged, and a classification may need an enriched incidence structure of cuts.

Question 19.5 (Recovering more than outer cofinality; source 03; open). Can an intrinsic invariant of the real closed field recover the least *length* of a fresh sign, rather than only outer cofinalities of such lengths?

Birthday length is not part of the field language. The least length of a fresh sign is $\beta(M, N)$ (remark 10.4), but β is defined through the ambient sign tree, so this does not answer the question.

Question 19.6 (Additional compatible structure; source 03; open). Which transported real forms can also be required to preserve a prescribed valuation structure, a surreal exponential, or a specified derivation on an old set-sized substructure?

The pure-field extension lemma fixes arbitrary set-sized field data but does not extend isomorphisms in these stronger languages. Homogeneous-universal results for suitable surreal H -fields [7] have additional hypotheses and should not be substituted without checking them.

Question 19.7 (Infinite families and relative real forms; sources 03, 08; open, finite version answered). Which external saturation spectra, and which infinite collections of regular thresholds or entire gap spectra, can be realized simultaneously by fixed fields of involutions of one SC^N that agree on $\mathbb{C}^N$ and have no set-sized cofinal subset?

Status. The finite-threshold version is answered by corollary 16.7 (source 03), and spectra arising from given inner models are realized by theorems 16.2 and 16.6. A classification of arbitrary real forms, or of infinite families, needs genuine iteration and spectrum analysis. This is also where the results continue the automorphism report's open real-form classification (remark 16.4).

Berenbeim's Question 2 (2020 notes) is answered in the precise set-complete interpretation of theorem 17.1, with the class-complete interpretation refuted under Global Choice (theorem 17.3). It is not an open question of the collection.

20 What this report does not claim

Every limitation, disclaimer and priority caveat of the four sources is kept. They are listed per source; items written for the merge are marked M.

Source 01

- 01.1 The principal spectrum and decoding results are proposed contributions with proofs, not certified historical priority; no Lean verification and no independent referee review.
- 01.2 Classical surreal facts, quantifier elimination and standard forcing facts are not claimed; several consequences are deliberately labelled consequences.
- 01.3 Berenbeim's Question 2 comes from 2020 research notes, not a peer-reviewed conjecture. The construction may be folklore in class forcing; it is not a claim that Boolean completions were previously unavailable, and general class-forcing completion theory predates the notes. The notes' broader suggestions about embeddings of universes are not settled.
- 01.4 External means computed in N , not in a larger metatheory; countable transitive models must be read internally. No unrestricted class truth predicate is assumed.

- 01.5 Field elementarity does not assert $M \preceq N$. The ground-bound lemma is not a covering property of a forcing, and the containing set may be much larger than the external set.
- 01.6 The decoder does not infer that an external branch is old because its nodes are old. Interval widths are not claimed to tend to zero.
- 01.7 Regularity of δ is an elementary minimal-length fact, not a new characterization of regular cardinals. The gap invariant excludes cuts with proper-class cofinality on one side.
- 01.8 Set-saturation of the pure field does not produce a class isomorphism; the nonisomorphism corollary requires set-sized images; proposition 12.3 is a consequence, not a new field theorem.
- 01.9 The distributivity equivalence is standard; statements are pair by pair, not about every generic extension of a poset unless so quantified; no preservation of larger cardinals is asserted.
- 01.10 A topology looking only at selected scales is a different structure and is not used.
- 01.11 The Boolean completion is not claimed for every class partial order nor claimed unique; set-completeness does not imply class-completeness; the negative result assumes Global Choice, with no optimality claim for that assumption.
- 01.12 No classification of other orders with the interval-method properties is claimed.
- 01.13 Exponential and differential expansions are not addressed. Arbitrary class functions and proper-class cuts are not claimed available in ZFC. Not a consistency proof for ZFC, GBC, measurables or the model configurations; Prikry is conditional on a measurable. Nothing is Lean-checked; typesetting is not verification.
- 01.14 Its three questions are recorded with their current status in section 19.
- 01.15 Its repository review was limited: the foundations report was read to line 180, the automorphism report only in its opening text, and code searches used a default-branch index. The repository was not modified or built. Literature searches were not exhaustive. Its novelty claim was: candidate contributions not located in the consulted literature or the stated repository review, subject to independent checking and further priority research.

Source 03

- 03.1 Historical novelty is not certified by an exhaustive literature review, peer review or formal verification; no named published conjecture is claimed solved.
- 03.2 The forcing theory of fresh functions is not renamed as a new invariant; quantifier elimination is standard; the collection's class back-and-forth, conjugacy/fixed-field criterion and countably cofinal real form are not claimed.
- 03.3 No set of all surreals, no class-object of all automorphisms, no class Zorn lemma and no general class-valued recursion; sizes are relative to N ; proper-class size in N is not proper-class size in every outside universe.
- 03.4 Principal cuts and cuts needing a proper-class presentation are not considered.
- 03.5 The invariant recovers a threshold and cofinality data only, not the inner model, the forcing or the sign lengths; equality of spectra is not proved to classify the fields.
- 03.6 $\kappa = \omega$ must not be substituted into the field-saturation clause. The invariant belongs to a pair of universes, not to a poset.

- 03.7 The $\text{Add}(\kappa, 1)$ computation does not compute the full gap spectrum or exclude larger gaps, and asserts nothing about preservation of higher cardinals. No Prikry property is reproved.
- 03.8 Set-saturation of a proper-class algebraically closed field is an established mechanism, not a novelty claim; class back-and-forth already exists in the collection; the inclusion is not the isomorphism Φ ; the conjugacy criterion is the transport mechanism, not the new invariant.
- 03.9 The conjugation structures are elementarily equivalent; Φ need not preserve simplicity, the natural valuation, the real exponential or a derivation, and no such compatibility is claimed.
- 03.10 Finite towers only: no infinite iteration, and no arbitrary proper class of invariants realized simultaneously.
- 03.11 Set-discreteness is classical and already in the collection; no convergence of an ordinary sequence to a missing point is asserted.
- 03.12 Tempting shortcuts are false or unproved: replacing a new ordinal-valued sequence by a binary one of the same length without a coding proof; interchanging ground and outer cofinalities; letting global cofinality control bounded gaps; inferring type preservation over new parameter sets from elementary equivalence; assuming a pure-field isomorphism preserves the real axis; inferring a full-class property from one set-sized workspace.
- 03.13 Not a Lean formalization; the repository was not rebuilt or axiom-audited; the Python program cannot test freshness, forcing, uncountable regularity or the class arguments; its proposed formalization order names proof tasks, not existing declarations.
- 03.14 Its questions are directions, not assertions that a published question remains open.
- 03.15 The repository comparison was focused, not exhaustive. Fischer–Koelbing–Wohofsky’s spectrum-realization questions are not resolved, and their convention for spectra of a forcing must not be conflated with cofinalities in a particular outer universe. The novelty statement is narrow; a specialist may identify an earlier theorem subsuming part or all of the arguments.
- 03.16 Its shipped `03-fresh-sign-gaps-PROOF-STATUS.md` lists what was established by argument, the established ingredients and seven boundaries, and states that no Lean proof, theorem-prover verification, forcing simulation or repository build was performed and that independent specialist review is still appropriate.

Source 04

- 04.1 Proposed contributions; no exhaustive priority claim, no solution of a named published problem, no Lean certification, no peer review.
- 04.2 Ehrlich’s absolutely saturated models and the universal H -fields of van den Dries–Ehrlich are precedents. External means internal to the specified outer universe.
- 04.3 Definability of M is a convenience. No claims about arbitrary externally countable models; class isomorphisms add Global Choice.
- 04.4 Not new: cut realization in **No**, quantifier elimination, saturation of algebraically closed fields with enough independent elements, conjugation naming a real closed fixed field, and many surcomplex automorphisms and real axes in the collection.

- 04.5 Birthday minimization over old candidates is not claimed absolute; “initial” means under simplicity, not numerically; the cover lemma is not the κ -cover property; no equivalence between $\eta_{\aleph_0}$ and $\aleph_0$ -saturation of a real closed field; no class-valued recursion.
- 04.6 No regularity assumption on κ ; cardinals are computed in the outer universe; its name for the first new length was local notation, not a claim to a new forcing invariant; regularity is a set-theoretic fact.
- 04.7 In the pure-field section the contrast is the contribution, not a classification of types.
- 04.8 Forcing statements quantify over every generic extension; one extension is not enough. The comparison table does not alter internal saturation.
- 04.9 The class back-and-forth is noncanonical and preserves no coordinates, valuation, real axis, simplicity or exponential. The several-spectra corollary concerns a given finite family; arbitrary spectra are not claimed producible.
- 04.10 Full-class fine-topology results are already in the collection.
- 04.11 The full text of Ehrlich’s 1989 paper was not accessible to it. No extension to exponential or differential languages; the Rangel–Mariano and Bertram questions are not resolved. Its novelty claim: proposed original results, with priority remaining to be independently established.
- 04.12 A code-search response marked incomplete was not treated as an exhaustive negative; the nonduplication statement is not a claim about later commits.
- 04.13 Its sensitive proof points (old elements are not an old set; a minimal fresh length is a limit; ordinal labels cannot be replaced by binary digits; class recursion is not silently assumed; external cardinal bounds belong to the outer universe; finite parameters need not give a finite type; forcing preservation quantifies over all generics; a pure field does not remember a chosen real form) and its finite-check limits are kept; no verified priority or solved named problem.

Source 08

- 08.1 Proposed contributions; classical theory credited; no Lean, no peer review, no named problem solved, no exhaustive priority claim.
- 08.2 The algebraically closed field statement is not a new principle; class back-and-forth is classical and the isomorphism is a corollary; exponentials, derivations and unrestricted automorphisms are not classified.
- 08.3 Global Choice is used only for the class isomorphisms. No countable set model and no set-valued field object is used.
- 08.4 The cover lemma is not a covering or approximation property; δ is undefined when $M = N$; ordinal sequences must not be replaced by binary ones.
- 08.5 The algebraically closed field argument is a classical instance; conjugation is not definable in the pure field.
- 08.6 Distributivity needs the uniform quantifier (lottery sum). Closure facts are standard; no GCH; no claim about preserving higher cardinals; Prikry properties are imported, not reproved.
- 08.7 Its real-form corollary does not classify involutions; the collection’s back-and-forth, conjugacy criterion and countably cofinal real form are prior work.

- 08.8 An omitted cut is not a Cauchy sequence, and no topological completion theorem is claimed. The ordered exponential field, the Berarducci–Mantova derivation and simplicity expansions are not treated. Nonisomorphism compares inner and ambient universes only; nothing is claimed about $\mathbf{No}^{M_1} \cong \mathbf{No}^{M_2}$.
- 08.9 Its questions are not claims about previously catalogued open questions.
- 08.10 No repository file was modified and no independent build was done; the finite checks do not establish transfinite claims.
- 08.11 It inspected a scan of Ehrlich’s 1989 paper; its repository and literature searches were limited, and absence of a search hit is weak evidence. Its warranted conclusion: a specific relative-universe theorem package not found in the inspected literature or repository material, with historical novelty subject to broader expert checking.

Added by the merge

- M.1 The merge adds no mathematical claim beyond the observation of remark 10.4; its other additions relate statements and reports, or restate a source’s statement in another source’s notation.
- M.2 The real-form results continue, and do not settle, the real-form classification question of the automorphism report.
- M.3 The relation to the Lean form of `found.thm:discrete` (remark 14.4) is information for formalization; nothing in this report is formalized.
- M.4 The sibling report is cited by directory and by what it is described to prove; this report makes no claim about its numbering or its other statements.
- M.5 The source manuscripts were not rebuilt for the merge; their page counts are as delivered.

This gives 15, 16, 13 and 11 items from sources 01, 03, 04 and 08, and 5 added by the merge: 60 in all.

A Provenance and repository comparison

A.1 Pinned review scope of the sources

All four sources were written against

VladimirReshetnikov/Surreal, commit 4f2645599121fa872c7104995e47f86f0382351f,

(“Index the newly placed drafts without extending review claims”), a commit that precedes the batch-20 rewrite of the automorphism report. They read the repository through a connector and did not modify or build it.

- Source 01 read docs/README.md, the opening 180 lines of the foundations report, the returned opening text of the automorphism report, and the root README; its scoped code searches used the connector’s default-branch index. It states that this is not a file-by-file audit.
- Source 03 read the main README, the catalogue, the foundations README, and the sections of the automorphism report on the unrestricted field, class back-and-forth and genuinely different real forms.

- Source 04 inspected the recursive file tree, the root description, the catalogue, and the foundations and automorphism sources, not every manuscript in full; a repository-wide code search returned an incomplete result that it did not treat as exhaustive.
- Source 08 read the root README, the documentation inventory, the automorphism report’s README, and selected source sections including the class back-and-forth, the involution conjugacy criterion and the countably cofinal real form.

A.2 Statements corrected against the current collection

1. *Searches for “forcing” and “saturation”.* Source 01 reports that searches for forcing, saturation and distributivity “returned no matches”; source 08 that searches for forcing and saturation “returned no hits”. At the pin, and now, the foundations report contains the phrase “saturation for small cuts” in its discussion of completeness and a subsection “ETR, class forcing, and two invalid moves” (found:sub:etr). Neither contains the results of this report, and both sources presented their searches as limited. Source 04’s statement that the foundations report’s forcing references did not supply the cross-universe spectrum is accurate.
2. *Report counts.* Source 01 describes docs/README.md as listing forty catalogued reports plus four newly placed drafts. That was the state at the pin. The collection now has 44 report directories, including this report and its sibling.
3. *The automorphism report.* It was rewritten after the pin (commit fb182ea, the batch-20 write), and its labels gained the prefix saut:. The sources’ descriptions of what they read in it (“opening text”, “the newly placed manuscript”, selected sections) refer to the pinned version. This report cites its current numbers and labels: Theorem 9.1 (saut:thm:acfhomogeneity), Section 10 (saut:sec:realforms), Theorem 10.2 (saut:thm:inequivalentrealform), Proposition B.1 (saut:prop:conjugacycriterion) and Section 14. Source 04’s remark that the automorphism source contained no matches for forcing, saturat or nonconjugate is true of the pinned text; the current text still has no saturat and no nonconjugate, and uses “forcing” once only in the unrelated sense of forcing an equation.
4. *A missed prior result.* Sources 03 and 08 correctly credit the collection with class back-and-forth, the fixed-field conjugacy criterion and the countably cofinal real form, but none of the four sources cites the set-cut interpolation criterion (present at the pin as prop:conjugacycriterion, now saut:prop:conjugacycriterion). Remark 16.4 records what it implies and what remains new.
5. *Formalized degeneracy.* Source 01’s statement that the internal degeneracy of set-indexed convergence is “discussed and formalized” in the repository is accurate: docs/FORMALIZATION.md marks the subset, convergent-net and Cauchy-net clauses of found:thm:discrete as proved, the last under HasSmallCutFillers (remark 14.4).

A.3 Literature boundary

Conway and Gonshor supply the construction and cut theory [1, 2]; van den Dries–Ehrlich with their erratum, and Bournez–Guilmant, the birthday-field background [3, 4, 5]; Marker the quantifier-elimination and saturation machinery [9]; Ehrlich’s absolutely saturated models the class-saturation background, whose terminology concerns all sets of the ambient universe and is not contradicted by the use of a larger universe [6]; van den Dries–Ehrlich’s homogeneous universal H -fields and Ehrlich–Kaplan’s surreal ordered exponential fields the richer structures [7, 8]; Jech, Prikry, Gitik and Benhamou the forcing facts [10, 11, 12, 13];

Fischer–Koelbing–Wohofsky the forcing notion of freshness [14]; Berenbeim the reverse-simplicity question and the suggestion to study surreals under forcing [15]; Holy and coauthors the class-forcing completion theory [16]. Rangel–Mariano and Bertram connect surreal constructions with set-theoretic foundations; their themes and questions are not claimed or resolved here [17, 18]. Each source’s targeted literature search did not locate the exact theorems; none is an exhaustive bibliography.

B Dependency and proof-obligation ledger

Result	Essential inputs	Boundary checked
Old arithmetic and elementarity	Absolute signs, prefix witnesses for old cuts, set-like Conway recursion; RCF/ACF QE	Field elementarity is not universe elementarity.
Ground-set containment	Birthday bounds; common ordinals; old bounded sign sets	The containing set need not be small.
One old side	Ground-set containment; ground cut theorem	Both sides external sets; one literally in M .
Regular first length	Minimal new sequence; old coding maps	Cardinality and cofinality computed in N ; M -regularity by persistence.
Interval tree and decoder	Set-length recursion; explicit splitting; two cuts at limits	Decoder total, with a default value, in M .
Absolute interval code	Per-input set recursion; absolute arithmetic and cuts	No set of all children; labels range over Ord.
Block codes	Two-bound cone identity; set-recursive decoding	Ordinal, not natural, sums; successor runs.
Main spectrum	Minimal new sequence; decoder; one-side lemma; RCF QE	Saturation equivalence only for uncountable κ .
First gap	Regularity; nested endpoints; one-side interpolation	Cofinalities in N with set witnesses.
Gap correspondence	Prefix convexity; simplest separators; constant-tail closure	Set-presented gaps only; no Global Choice.
Least gap	Old-object coding; terminal-plus blocks	Sizes computed in N .
First new birthday	Characteristic signs; minimality; ordinal coding in M	β is not a birthday of a given number.
ω -saturation	Countable syntax over finite old parameters; rational cuts	No new reals is not no new countable sequences.
Pure surcomplex saturation	Ground envelopes; algebraic closures of set fields; ACF QE	No class-sized transcendence basis or global isomorphism.
Conjugation spectrum	Fixed field; order by nonzero squares; coordinates	Naming i adds one parameter.
Bounded fragments	Birthday cut bound; sign-cone lemma	Regularity in N separate from equality of short signs.
External Cauchy rigidity	Old positive lower bounds	Fine surreal radii; no cut filling needed.
Closed, nowhere dense	Open cones; old radii below new ones	Needs $M \neq N$ for nowhere-denseness.
Class isomorphism, real forms	Global Choice; set-stage back-and-forth; gap invariance	No valuation, simplicity or exponential preserved.
Set-complete Boolean algebra	Cylinder maps; common levels for set families	Least-level set pairs, not class objects.

Result	Essential inputs	Boundary checked
No class-complete algebra	Global Choice; proper-class antichain; diagonal subclass	Stronger completeness, extra axiom.

The ledger records the mathematical checks made in assembling the arguments. It is not a proof-assistant log.

C Finite checks

Three sources shipped standard-library Python programs; source 01 shipped none. They check finite sign comparisons and interval formulas. There are no finite fresh signs, since a finite sequence of old entries is always old, so none of them tests freshness, forcing, regularity, saturation or any class argument.

- `code/03-fresh-sign-gaps-check_finite_signs.py` checks first-disagreement comparison, convexity of finite prefix cones and the terminal-plus block code with its endpoint ordering, for value lists of length at most four with entries in $\{0, 1, 2, 3\}$ and cones among signs of length at most six. Recorded run (`data/03-fresh-sign-gaps-finite_checks.json`): 127 sign strings, 16,129 ordered-pair comparisons, 127 cone checks, 340 block-code cases and 1,744 nested endpoint pairs, all passing. For the list $(2, 0, 1)$ the code is $\text{code}^+ = +++-+-++-$, with endpoint lengths $(2, 3)$, $(4, 5)$, $(7, 8)$ and values $2 < \frac{5}{2} < \frac{43}{16} < \frac{173}{64} < \frac{87}{32} < \frac{11}{4} < 3$.
- `code/04-universe-saturation-verify_finite_models.py` checks, in exact rational arithmetic, the successor interval formulas (5.4), the closed formulas of proposition 5.5, separation of children with finite labels, unique finite decoding, and the cut/prefix equivalence of lemma 2.2 for finite strings. Recorded run (`data/04-universe-saturation-finite_check_results.txt`): 1,365 interval nodes, 20,475 child pairs, 1,364 decodings, 65,025 cut/prefix comparisons.
- `code/08-forcing-saturation-verify_sign_code.py` checks sign order against dyadic values, the two-bound identity of lemma 5.6, decoding, nesting and finite cone intersections for $\text{code}(f)$. Recorded run (`data/08-forcing-saturation-verification_results.txt`): 304,000 checks. For $(2, 0, 1)$ the code is $++-+ -+ +-+$ with value $\frac{219}{128}$ and intervals $(\frac{3}{2}, 2) \supset (\frac{13}{8}, \frac{7}{4}) \supset (\frac{109}{64}, \frac{55}{32})$.

All three suites were rerun for this merge on a copy of the directory, with Python 3.14; each output agrees with its recorded file except for line endings on Windows. Commands are given in the README.

References

- [1] J. H. Conway, *On Numbers and Games*, 2nd ed., A K Peters, 2001 (first edition, Academic Press, 1976).
- [2] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [3] L. van den Dries and P. Ehrlich, “Fields of surreal numbers and exponentiation,” *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>.
- [4] L. van den Dries and P. Ehrlich, “Erratum to ‘Fields of surreal numbers and exponentiation’,” *Fundamenta Mathematicae* **168** (2001), 295–297. The corrected statements are also restated in [5, Theorems 1.1–1.2 and 3.3].
- [5] O. Bournez and Q. Guilmant, “Surreal fields stable under exponential and logarithmic functions,” arXiv:2201.08199, 2022. <https://arxiv.org/abs/2201.08199>.
- [6] P. Ehrlich, “Absolutely saturated models,” *Fundamenta Mathematicae* **133** (1989), 39–46. <https://doi.org/10.4064/fm-133-1-39-46>.
- [7] L. van den Dries and P. Ehrlich, *Homogeneous universal H-fields*, arXiv:1807.08861, 2018. <https://arxiv.org/abs/1807.08861>.
- [8] P. Ehrlich and E. Kaplan, *Surreal ordered exponential fields*, arXiv:2002.07739, version 3, 2021. <https://arxiv.org/abs/2002.07739>.
- [9] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics 217, Springer, 2002. <https://doi.org/10.1007/b98860>.
- [10] T. Jech, *Set Theory: The Third Millennium Edition, Revised and Expanded*, Springer Monographs in Mathematics, Springer, 2003. <https://doi.org/10.1007/3-540-44761-X>.
- [11] K. L. Prikry, “Changing measurable into accessible cardinals,” *Dissertationes Mathematicae (Rozprawy Matematyczne)* **68** (1970), 55 pp.
- [12] M. Gitik, “Prikry-type forcings,” in M. Foreman and A. Kanamori (eds.), *Handbook of Set Theory*, Springer, 2010, 1351–1447. https://doi.org/10.1007/978-1-4020-5764-9_17.
- [13] T. Benhamou, “Prikry forcing and tree Prikry forcing of various filters,” arXiv:1801.04424, 2018. <https://arxiv.org/abs/1801.04424>.
- [14] V. Fischer, M. Koelbing and W. Wohofsky, “Fresh function spectra,” *Annals of Pure and Applied Logic* **174** (2023), no. 9, 103300. <https://doi.org/10.1016/j.apal.2023.103300>.
- [15] A. Berenbeim, *Review of Surreal Number Construction and Forcing Construction*, research notes dated 8 April 2020, 35 pp. Question 2, p. 32; proposed further directions, p. 34. Listed by the author as incomplete notes. <https://www.berenbeim.com/papers>.
- [16] P. Holy, R. Krapf, P. Lücke, A. Njegomir and P. Schlicht, “Class forcing, the forcing theorem and Boolean completions,” arXiv:1710.10820, 2017. <https://arxiv.org/abs/1710.10820>.
- [17] D. Rocha Rangel and H. L. Mariano, *An algebraic (set) theory of surreal numbers, I*, arXiv:1911.12726, 2019. <https://arxiv.org/abs/1911.12726>.

- [18] W. Bertram, *On Conway's Numbers and Games, the Von Neumann Universe, and Pure Set Theory*, arXiv:2501.04412, version 2, 2025. <https://arxiv.org/abs/2501.04412v2>.
- [19] *Automorphisms of the Surcomplex Numbers*, report in VladimirReshetnikov/Surreal, docs/surcomplex/surcomplex-field-automorphisms/, label prefix saut:. The sources read its pinned version at 4f26455.